

Learning Optimal Liquidation with Closing Auctions

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September 7, 2026

Abstract

We study liquidation when continuous trading is followed by a closing auction. The trader sells through a limit-order book, then submits signed auction schedules to adjust the remaining inventory. A projected clearing signal and intermediate auction feedback connect the two phases. We compare DQN with projected DDPG, TD3 and SAC policies on a common executable grid, using synthetic rough-Heston prices and historical midprice paths within a simulated market. Policies are selected and evaluated by inventory-penalized implementation shortfall, separately from the weighted training objective. They improve on stylized AS and TWAP references, and matched synthetic comparisons support useful auction access for all four learners. Dense auction credit improves learning. The clearing forecast is informative, but its incremental decision value depends on the learner; added forecast-relative preferences show no established marginal benefit when learning conditioning is matched. These results concern the stated simulator and calibration, including reused historical test dates, rather than real-exchange profitability.

Key words: optimal liquidation, auction trading, reinforcement learning, Deep Q-Learning

1 Introduction

1.1 Reinforcement Learning for Electronic Trading and Liquidation

Market making is a cornerstone of modern electronic financial markets, providing liquidity by continuously posting buy and sell quotes while managing inventory and adverse selection risk with different participants having diverse objectives. Since the seminal work of Avellaneda and Stoikov [4] and its extension to explicit solutions in [31], optimal market making has been studied extensively through stochastic control frameworks, leading to tractable strategies that balance expected profit against inventory risk under stylized assumptions on order flow dynamics and price evolution. We refer to [16] for a review of the literature on market making and high-frequency trading and to [5, 6] for recent advances in the topic. These models, however, typically rely on parametric assumptions that are difficult to validate empirically and may fail to adapt to the non-stationarity and strategic complexity of real-world markets.

The rapid growth of electronic trading and data availability combined with AI’s growing influence in the financial industry has motivated the use of reinforcement learning as a flexible, data-driven alternative to classical control methods. Reinforcement learning allows a market maker to learn optimal

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quoting policies directly from interaction with the market, without requiring full knowledge of the underlying dynamics. Recent studies have demonstrated the promise of RL in market making and related problems, showing improved adaptability to complex market conditions, latent regimes, and evolving order flow patterns. Nevmyvaka et al. [54] combine Q-learning and dynamic programming for optimized trade execution using limit-order-book information. Related reinforcement-learning studies include [7, 55, 27, 30, 36, 41], with regret analyses in [17, 13]. Related predictive work uses deep neural networks to model future limit-order-book states [59]; this is distinct from learning trading policies by reward optimization. For broader treatments, see the reinforcement-learning survey [33] and the volume on machine learning and data sciences for financial markets [14]; [18] is an editorial introducing a special issue on machine learning in finance.

While the notion of market making is an active topic of research including new developments with reinforcement learning techniques, one aspect remains insufficiently explored in the RL-based market making literature. At the end of the day, most exchanges are closing the market by triggering a closing auction, which plays a fundamental role in price discovery and liquidity provision in modern equity markets. Closing auctions concentrate a significant fraction of daily trading volume and often exhibit dynamics that differ markedly from those observed during continuous trading. Inventory held into the closing auction can be liquidated at a single clearing price, but doing so exposes the market maker to auction-specific price impact, imbalance risk, and strategic interactions. Traditional market making models typically ignore the auction or treat the end-of-day liquidation in an ad-hoc manner, while most reinforcement learning approaches focus exclusively on continuous limit order book trading.

We consider a trader who begins with a positive inventory and submits sell limit orders during a continuous CLOB phase. At the auction opening, the control set is enlarged to signed auction schedules. The agent learns a feature-based policy without knowing the transition law of the exogenous order flow, while the exchange clearing mechanism itself is specified by the model.

1.2 On the Importance of (Closing) Auctions

Early work on financial auctions includes Kyle's model [40]. In this article, Kyle provides the first tractable equilibrium model of a continuous auction with asymmetric information. It explains how prices aggregate private information through order flow, introducing a measure of market depth and price impact. This paper became the benchmark for analyzing liquidity, price discovery, and strategic trading in modern auction-based financial markets. While auction markets have been well-investigated in the economics community for discrete-time models (see for example [47, 62, 50]) and have known a growing interest, especially since the work of the Nobel laureate Paul Milgrom, they still represent a challenging problem both at the high frequency-level and for continuous-time frameworks, and have been identified as one of the most challenging questions in financial engineering in [15, Section 5.1].

The big picture of the trading session considered in this paper is the following. During continuous trading, the agent solves a one-sided end-of-day liquidation problem: she can only sell and her inventory remains nonnegative. During the closing call, the admissible action set is enlarged to permit two-sided auction schedules, so that terminal inventory may lie on either side of zero. The auction phase is therefore not constrained to monotone liquidation. Deviations from zero terminal inventory are discouraged through the terminal inventory penalty rather than prohibited by a hard position constraint. At the end of the auction phase, known as the clearing time, a clearing price is set by the exchange to ensure as much transaction as possible to clear the market and trade the asset. This closing auction plays a fundamental role in market liquidation and empowers efficiency

of price discovery as explained in [38, 56]. We also refer to [48] for an overview of auction mechanism.

On the one hand, auctions successfully fix mechanical flaws of limit order book like correlation breakdown: albeit their mechanism is distinct from the end-of-day inventory-management role studied here, frequent batch auctions have been proposed to mitigate the speed race associated with serial continuous trading, as explained in [11]. On the other hand, unlike a LOB trading, the key challenge of the auction phase is to set an efficient (clearing) price to trade the asset, given the different operations of market participants, known as the price discovery mechanism [44, 8, 9]. The benefit of auctions for market quality, as reducing the spread between the clearing price and the efficient price of a risk asset has been investigated in the recent literature, for example [21, 37, 19, 58, 34] by proposing incentives and optimal fees scheme to mitigate auctions' flaws [45, 46].

Finally, we see in many markets nowadays that volumes at the closing auction have increased significantly in recent years. Ignoring this stage is therefore problematic when modeling market making. Furthermore, high-frequency market makers aim at ending each trading day with a flat position: it is likely they will use the auction to close their position if they need to. This underlines the importance of considering closing auctions in an end-of-day market making or liquidation framework.

1.3 Methodology, Contributions and Financial Insights

The central question is whether a trader can use information about a future closing auction to balance continuous liquidation with terminal inventory management. We connect sell-only CLOB execution to a closing call that accepts signed schedules. The trader must decide both how much inventory to carry into the call and how to adjust that position before clearing. A projected clearing signal links the phases, while intermediate feedback assigns learning credit before auction cash settles. Terminal positions may remain long or short; the inventory penalty discourages, rather than prohibits, residual exposure.

We formulate a single-agent, feature-based reinforcement-learning problem [60], connecting financial applications in allocation, execution and market making [53, 26, 39, 55, 5] to explicit closing-call control. DQN [64, 51, 23, 52, 61] is the discrete-action baseline; DDPG, TD3 and SAC [42, 24, 32] propose continuous controls that are projected onto the same executable grid. The phase-specific architectures, boundary continuation and conditioning are specified in Section 4. These are feature-based implementations, not a claim that the reduced observation is Markov sufficient.

The headline policies train on the weighted preference objective J . Potential and reference adjustments are used only for learning; checkpoint selection and policy comparisons instead minimize inventory-penalized implementation shortfall, excluding fictive rewards. Our references are a stylized AS policy adapted from [4, 31] and a time-weighted average price policy. The stylized benchmarks provide a reproducible analytical reference for evaluating the proposed framework; comparisons are conditional on the stated heuristic calibration and are not interpreted as establishing superiority over optimally calibrated execution models.

The completed study addresses three empirical questions. First, do the learned policies use the closing call to improve liquidation and inventory management? Second, does forecast accuracy translate into incremental decision value? Third, does dense feedback help learn from delayed auction outcomes, separately from the additional preferences in J ? Across the tested settings, headline policies improve inventory-penalized shortfall relative to the references, although this does not imply uniformly better ordinary shortfall or a universal algorithm ranking. Realized auction contributions and the matched retrained-access comparison support useful auction participation, with positive access-effect intervals for all four learners. Matched synthetic comparisons show benefits from dense

auction credit, learner-dependent incremental H-feature value, and no established marginal preference benefit once conditioning is held fixed. The negative anchoring results and adverse historical outcomes are retained in Section 6.4.

Outline. Section 2 specifies the market, clearing rule and projected clearing signal. Section 3 defines admissible controls, the training objective and economic shortfall criteria. Section 4 describes the learners, followed by the reference policies in Section 5. Section 6 presents the experimental protocol, simulator calibration and results, distinguishing the main campaign from its two synthetic follow-ups. Section 7 concludes.

2 Market Model

2.1 Mathematical Framework and Trading Phases

Along this work, we fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, named the market, where Ω represents all the possible market configurations, $\mathcal{F}$ is a σ -algebra denoting the available information and $\mathbb{P}$ is the market probability. We consider a financial asset traded on the financial market along a trading session with price evolving randomly. We divide the trading period into two phases: a continuous phase on a limit order book with market makers and takers and an auction phase, seen as a usual closing auction. We fix two deterministic times $0 < \tau^{\text{op}} < \tau^{\text{cl}}$ representing respectively the opening and the closing of the auction phase (in minutes) and denote by $h = \tau^{\text{cl}} - \tau^{\text{op}}$ the auction duration. We assume that both $(\tau^{\text{op}}, \tau^{\text{cl}}) \in \mathbb{N}^2$ and h is a positive integer, with $\tau^{\text{op}} \geq 2$. Therefore, the trading horizon is divided into a continuous phase $[0, \tau^{\text{op}})$, in which the trader interacts with the central limit-order book (CLOB) and a fixed length h closing auction $[\tau^{\text{op}}, \tau^{\text{cl}}]$. The agent acts on a realized decision grid $0 = t_0 < \dots < t_n < \tau^{\text{op}} = t_{n+1} < \dots < t_m < \tau^{\text{cl}} = t_{m+1}$. During the continuous phase, the decision times $t_0, \dots, t_n$ are random times generated by the exogenous market-order processes as in Definition 3.1. Consequently, n is random. The decision clock is nevertheless control-independent: conditional on the same realization of the exogenous market processes, the sequence $(t_i)_{0 \leq i \leq n}$ is the same for every admissible policy. During the auction phase, decisions are made at deterministic one-unit intervals such that $t_{n+1+\ell} = \tau^{\text{op}} + \ell$ for $\ell \in \llbracket 0, h \rrbracket$, where $t_{n+1+h} = \tau^{\text{cl}}$ is terminal. In particular, $m = n + h$. The initial time t_0 is the start of the considered trading session, but need not be the actual opening time of the market.

We denote by I_t the trader's inventory at time t . This inventory is positive (resp. negative) for long position (resp. short positions) with respect to the traded risky asset.

We denote by $\alpha > 0$ the tick size of the asset, fixed by the exchange. We will consider three types of market participants in this study:

- a strategic trader, called the agent, who liquidates inventory through sell limit orders during continuous trading and submits signed schedules during the closing call,
- exogenous market makers, fixing limit orders and providing liquidity during the continuous trading session on both sides of the LOB and proposing limit prices during the auction session,
- exogenous market takers. These participants submit aggressive market orders during both the CLOB and the auction phases to buy or sell the asset.

While we will focus on the optimization of the agent along the day, we use the term exogenous to emphasize the fact that other market makers and takers' optimizations are not considered here. In

fact, we consider a single-agent decision problem against fixed background policies. All “exogenous” market participants are represented by a fixed stochastic data-generating mechanism which will be introduced later. This does not mean these participants have no strategic motives in an actual market, but that their individual optimization problems and endogenous responses to the agent are outside the model. We are therefore solving a best-response problem against the specified background policy, not a market-equilibrium problem. The two trading phases are non-overlapping by construction. We view this as a stylized sequential, Xetra-like chronology rather than as a complete model of a US closing process in which auction order collection may overlap with continuous trading. In the historical-input experiments, US equity data determine only the mid-price paths; order flow, auction updates and clearing remain synthetic. We now turn to the details of the trading period $[0, \tau^{\text{op}}) \cup [\tau^{\text{op}}, \tau^{\text{cl}}]$ composed by continuous trading activities on the CLOB followed by a closing auction.

2.1.1 Trading During the Continuous Phase $[0, \tau^{\text{op}})$

During the continuous phase on the LOB, we assume that each market participant observes the mid price $S_t^{\text{mid}} = \alpha k_t^{\text{mid}}$ at any time $t < \tau^{\text{op}}$, where $k_t^{\text{mid}} \in \mathbb{N}$ represents the number of ticks at which the mid price is priced. Exogenous market takers submit market orders on both sides of the market and consume liquidity. The number of market takers arriving on the side $\zeta \in \{+, -\}$ follows a counting process denoted by N_t^ζ , where $\zeta = +$ denotes the ask side, and $\zeta = -$ the bid side. In other words, N_t^ζ market takers have arrived on the side ζ (e.g., if $\zeta = +$, then N^+ corresponds to buying market takers, consuming liquidity on the sell side). We assume that each market taker $i \leq N_t^\zeta$ submits a market order with volume $\nu_t^{\zeta, i}$ on the ζ side of the CLOB at time t .

During the continuous phase, the agent acts as a one-sided liquidator and submits only sell limit orders. She submits limit orders characterized by a limit price denoted by $S_t^\bullet = \alpha k_t$ at time t where k_t denotes the number of ticks chosen at time t to price the asset, and a proposed volume v_t . The agent therefore submits an order characterized by the pair (k_t, v_t) at time t on the limit order book. We assume that the agent is selling her inventory I on the LOB during the continuous phase. This order is thus a limit order on $v_t \leq I_t$ shares at price $S_t^\bullet = \alpha k_t$.

Remark 2.1. Note that $\delta_t := k_t - k_t^{\text{mid}}$ represents the number of ticks between the sell limit order proposed by the agent and the mid price, seen as the ask-spread of the agent.

The liquidity provided by exogenous market makers at price level $S_t^{\zeta, j} = \alpha k_t^{\zeta, j}$, where $k_t^{\zeta, j} = k_t^{\text{mid}} + \zeta j$ for $j \in \mathbb{N}^*$ and $\zeta \in \{+, -\}$ is given by the volume $V_t^{\zeta, j}$ at any time t . We assume that there exists $L_\infty > 0$ such that the depth of the order book on side ζ is given by

$$L_t^\zeta = \sup\{j \in \llbracket 1, L_\infty \rrbracket : V_t^{\zeta, j} > 0\}$$

with the convention that $\sup \emptyset = 0$.

Assumption 1. Conditional on the price level selected by the agent, her order has priority over exogenous orders resting at the same price level.

To abstract from queue-position dynamics and latency competition, we adopted the favorable execution convention that the agent has execution priority with respect to other participants.

Remark 2.2. Based on the random arrivals of market takers, there is (at least partial) execution on the order of the agent if given that a market taker arrives at time t the following condition is

satisfied

$$\sum_{i=N_t^++1}^{N_t^+} \nu_t^{+,i} > \sum_{j=1}^{\delta_t-1} V_t^{+,j}.$$

Note that as soon as the buying volume of market takers reaches selling index k_t of the agent, her order gets (at least partially) executed. The whole remark holds under the premise that the agent has a live sell order with positive remaining volume.

For $i \in \llbracket 0, n \rrbracket$, let E_{t_i} denote the total number of shares executed over $(t_i, t_{i+1}]$ from the order submitted by the agent at t_i . This quantity is observed at t_{i+1} and is given by

$$\forall i \in \llbracket 0, n-1 \rrbracket, \quad E_{t_i} = \max \left(0, \min \left(v_{t_i}, \sum_{j=N_{t_i}^++1}^{N_{t_{i+1}}^+} \nu_{t_{i+1}}^{+,j} - \sum_{j=1}^{\delta_{t_i}-1} V_{t_i}^{+,j} \right) \right)$$

and

$$E_{t_n} = \max \left(0, \min \left(v_{t_n}, \sum_{j=N_{t_n}^++1}^{N_{\tau^{\text{op}}}^+} \nu_{\tau^{\text{op}}}^{+,j} - \sum_{j=1}^{\delta_{t_n}-1} V_{t_n}^{+,j} \right) \right).$$

We recall that E_{t_i} is a random variable since v_{t_i} and $V_{t_i}^{+,j}$ are random. The inventory of the agent between t_i and t_{i+1} is $I_{t_{i+1}} = I_{t_i} - E_{t_i}$.

Motivated by the reinforcement learning approach, we assume that the agent will trade during this session until the deterministic physical time $t_n = \tau^{\text{op}} - 1$ before the market switches to the closing auction phase, where n is the final CLOB decision index; hence $t_0, \dots, t_n$ comprise $n+1$ CLOB decisions.

Every order submitted by the agent at a CLOB decision time t_i is good for the single decision interval $(t_i, t_{i+1}]$. Any unexecuted remainder expires without a cancellation fee at t_{i+1} . In particular, the order submitted at t_n is active on $(t_n, \tau^{\text{op}}]$ and any remaining quantity expires at τ^{op} . It is not transferred to the auction.

2.1.2 Trading During the Auction Phase $[\tau^{\text{op}}, \tau^{\text{cl}})$

At time τ^{op} , the system transitions to an auction, opened by the exchange. Similarly to [19, 45] and motivated by the reinforcement learning approach with the Markov Decision Process modeling the agent interacting with the market, we assume that the agent is setting bids at fixed deterministic times during the auction.

Assumption 2. *The agent bids along the auction at discrete times $\tau^{\text{op}} = t_{n+1} < \dots < t_m < \tau^{\text{cl}}$.*

The remaining inventory $I_{\tau^{\text{op}}}$ can be adjusted through the auction; it need not be fully liquidated. More precisely, for all $t \in \{t_{n+1}, \dots, t_m\}$, the agent observes exogenous market and limit orders arriving on the auction. Market orders are composed by a certain volume to be bought/sold regardless of the clearing price that is set at the clearing time by the exchange, while limit orders are set along the auction through a supply function (functional volume to sell/buy above/below a certain price). Exogenous cancellations are subject to the proposal and validity rules below; the designated persistent exogenous schedule cannot be canceled. The agent may submit a schedule and, when admissible, cancel all her prior live schedules. In this sense, the agent reacts to the

environment (since she posts her order after seeing the orders of other market participants). After her final action at the terminal trading time $t_m < \tau^{\text{cl}}$, the system transitions to a final state that will allow to compute the clearing price and the exchanged volume, thereby allowing computation of the terminal reward of the agent. At t_m , the exogenous update precedes the agent's single final action. The resulting book is then frozen: no participant acts and no further event is processed on $(t_m, \tau^{\text{cl}}]$. Similarly to [21, 37, 46] we assume that the agent submits a linear supply curve to the auction stated in the following assumption.

Remark 2.3. *The ordering of moves immediately before clearing is a maintained stylization of the model, as the agent has a final reaction opportunity. The numerical conclusions are conditional on this ordering; no separate cancellation effect is identified.*

Assumption 3. *The agent has a linear supply curve $\Sigma_t : p \in \mathbb{R}_+ \mapsto K_t^a(p - S_t^a)$ for all $t \geq \tau^{\text{op}}$, where $K_t^a \geq 0$ and $S_t^a \in \alpha\mathbb{N}$.*

At each $t_j \in \{t_{n+1}, \dots, t_m\}$ the agent controls $K_{t_j}^a \geq 0$ and $S_{t_j}^a \in \alpha\mathbb{N}$ so that $\Sigma_{t_j}(p)$ represents net supply, that is the number of shares the agent is willing to sell at price p . If $\Sigma_{t_j}(p) \leq 0$ the agent is willing to buy at price p or below and conversely if $\Sigma_{t_j}(p) \geq 0$ the agent is willing to sell at price p or above. Allowing both signs enlarges the auction control set beyond monotone liquidation. The objective nevertheless remains end-of-day inventory reduction, encouraged through the terminal inventory penalty.

At auction decision time t_j , canceling past bids costs $d_{t_j} := (j - n - 1)d = (t_j - \tau^{\text{op}})d$, for $j \in \{n+1, \dots, m\}$ and $d > 0$. The agent can cancel if and only if at least one prior strategic schedule is live, and if doing so, removes every prior live schedule. Let $c_{t_j} \in \{0, 1\}$ where $c_{t_j} = 1$ if and only if, at time t_j , the agent cancels all orders submitted prior to time t_j . In particular, at the opening of the auction, we set $c_{t_{n+1}} = 0$. We track these actions through a variable θ_{t_j} , with $\theta_{t_n} = \mathbf{0}$ and $\theta_{t_{n+1}} = \mathbf{0}$, where $\mathbf{0} \in \mathbb{R}^h$ is the $m - n$ -vector with 0 components. We recursively define $\theta_{t_j} = \theta_{t_{j-1}} \vee c_{t_{j-1}} \sum_{k=1}^{j-n-2} \mathbf{e}_k \mathbf{1}_{\{K_{t_{n+k}}^a > 0\}}$ for $j \in \llbracket n+2, m+1 \rrbracket$ where $\mathbf{e}_k = (\delta_{i,k})_{1 \leq i \leq m-n} \in \mathbb{R}^{m-n}$. By definition, $\theta_{t_j}^{(s)}$ indicates whether the order submitted at time t_{n+s} for $s \in \{1, \dots, j - n - 2\}$ has been canceled prior to time t_j . More precisely, $\theta_t^{(s)} = 1$ if and only if the positive-slope schedule submitted in slot s has already been canceled, and $\theta_t^{(s)} = 0$ if that slot contained no order. Therefore, each schedule can be canceled at most once.

Remark 2.4. *In this formulation, the cancellation fee $d_{t_j} c_{t_j}$ increases linearly with elapsed auction time $t_j - \tau^{\text{op}}$ when cancellation occurs. We interpret this cost not as a strictly cumulative fee on historical orders – which formally could be subjected to redundant cancellations – but rather as a structural mechanism by the exchange to penalize “illusory liquidity”: it is an economic modeling device designed to discourage repeated provision and withdrawal of liquidity. The “fictive” auction reward introduced below is instead an algorithmic shaping term. It is not an exchange rebate and is not included in the economic PnL criterion.*

At each trading time $t \in \{t_{n+1}, \dots, t_m\}$, the agent submits an order (K_t^a, S_t^a, c_t) to the market. Note that with the definition above, c_t has no impact on the order (K_t^a, S_t^a) , only on orders sent at time strictly before t . The inventory of the agent remains frozen during the auction, i.e. $I_t = I_{\tau^{\text{op}}}$ for $t < \tau^{\text{cl}}$.

During the auction, we suppose in addition that both exogenous market makers and market takers are present in the auction. At each trading time t , the number of bids sent by exogenous market makers is denoted by M_t . Each bid sent by these actors is a limit order, each with volume $g_{i,t}(p)$,

which is the supply schedule of the i -th bid offer present at time t for $i \leq M_t$. Each exogenous $g_{r,t}$ is a signed net-supply schedule: positive values represent supply and negative values represent demand.

Market takers submit market orders in the auction. We retain the cumulative buy and sell arrival counters N_t^+ and N_t^- from the continuous phase, without resetting them. For $\zeta \in \{+, -\}$, $N_{\tau_{\text{op}}-}^\zeta$ is the count immediately before the opening auction update. Auction orders present at time t have indices $N_{\tau_{\text{op}}-}^\zeta < i \leq N_t^\zeta$, and $\nu_t^{\zeta,i}$ denotes their current volume. Each accepted auction arrival increments the corresponding counter by one and receives that index; a rejected proposal receives no index. At time t_j , an accepted cancellation of a previously active auction order is recorded as

$$\nu_{t_j}^{\zeta,i} = 0, \quad N_{\tau_{\text{op}}-}^\zeta < i \leq N_{t_{j-1}}^\zeta, \quad j \in \{n+2, \dots, m\}.$$

Earlier book snapshots remain unchanged. If an existing order is not canceled, its current volume is copied from $\nu_{t_{j-1}}^{\zeta,i}$ to $\nu_{t_j}^{\zeta,i}$. New accepted orders have indices $N_{t_{j-1}}^\zeta < i \leq N_{t_j}^\zeta$ for $j \geq n+2$, and $N_{\tau_{\text{op}}-}^\zeta < i \leq N_{\tau_{\text{op}}}^\zeta$ at opening. A cancellation of a newly accepted order in the same exogenous update is likewise recorded only in the current snapshot, as permitted by Algorithm 3. Thus cumulative arrival indices are nondecreasing, whereas cancellations can reduce active order count and volume. All auction market-order sums below exclude the CLOB indices $i \leq N_{\tau_{\text{op}}-}^\zeta$.

2.1.3 Clearing Price Rule and Estimation Along the Auction

The market makers and takers first send a last order in the auction, then the agent sends a final limit order and/or cancels past ones. After time t_m , the system moves into a final stage to set the clearing price of the auction. At time $t = \tau^{\text{cl}}$, the revised rule selects a uniform tick price $S_{\tau^{\text{cl}}}^{\text{cl}}$ that maximizes matched volume, then allocates the long side pro rata. This pro-rata allocation rule is a stylized modeling assumption. It preserves execution balance and treats participants proportionally without introducing order-level priority information absent from the model. First compute the continuous root p_{m+1}^* of aggregate net supply; this root, not necessarily the executable tick price, solves the equation

$$\sum_{i=1}^{M_{t_m}} g_{i,t_m}(p) + \sum_{s=n+1}^m \left(1 - \theta_{t_{m+1}}^{(s-n)}\right) K_{t_s}^a(p - S_{t_s}^a) - \sum_{\zeta \in \{+,-\}} \sum_{i=N_{\tau_{\text{op}}-}^\zeta+1}^{N_{t_m}^\zeta} \zeta \nu_{t_m}^{\zeta,i} = 0, \quad \text{for } p \in \mathbb{R}_+. \quad (1)$$

Along the auction, we assume that the agent computes the projected clearing price by solving the equation

$$\sum_{i=1}^{M_{t_{j-1}}} g_{i,t_{j-1}}(p) + \sum_{s=n+1}^{j-1} \left(1 - \theta_{t_j}^{(s-n)}\right) K_{t_s}^a(p - S_{t_s}^a) - \sum_{\zeta \in \{+,-\}} \sum_{i=N_{\tau_{\text{op}}-}^\zeta+1}^{N_{t_{j-1}}^\zeta} \zeta \nu_{t_{j-1}}^{\zeta,i} = 0, \quad \text{for } p \in \mathbb{R}_+ \quad (2)$$

which is the clearing price equation were the auction to close at time $t_j \in \{t_{n+2}, \dots, t_{m+1}\}$, and then applying the volume-maximizing tick projection to obtain an element of $\alpha\mathbb{N}$. The estimation at time $t_j = t_{m+1} = \tau^{\text{cl}}$ corresponds to solving (1) fitting with the exchange clearing rule.

Remark 2.5. For $j \geq n+2$, the label $H_{t_j}^{\text{cl}}$ denotes the indicative price available to the agent at decision time t_j obtained by solving (2) at time t_j and applying the volume-maximizing tick projection. It is computed from the exogenous auction book recorded at t_{j-1} and from all still-active agent schedules submitted up to and including t_{j-1} . Consequently, the time index of $H_{t_j}^{\text{cl}}$ records when the estimate becomes available, rather than the date of the underlying book snapshot.

We now provide sufficient conditions ensuring existence of a solution to (2). First, note that Equation (2) can be written as $\Phi_j(p) = 0$ where

$$\Phi_j(p) = \sum_{i=1}^{M_{t_{j-1}}} g_{i,t_{j-1}}(p) + A_j p - B_j - C_j$$

with

$$A_j = \sum_{s=n+1}^{j-1} \left(1 - \theta_{t_j}^{(s-n)}\right) K_{t_s}^a, \quad B_j = \sum_{s=n+1}^{j-1} \left(1 - \theta_{t_j}^{(s-n)}\right) K_{t_s}^a S_{t_s}^a, \quad C_j = \sum_{\zeta \in \{+, -\}} \sum_{i=N_{\tau_{\text{op}}-}^{\zeta}+1}^{N_{t_{j-1}}^{\zeta}} \zeta \nu_{t_{j-1}}^{\zeta, i}.$$

Theorem 2.1 (Existence of a unique (estimated) clearing price). *Let $t_j \in \{t_{n+2}, \dots, t_{m+1}\}$. Assume that $\lim_{p \rightarrow +\infty} g_{i,t_{j-1}}(p) = +\infty$ and $p \mapsto g_{i,t_{j-1}}(p)$ is continuous and strictly increasing for any $i \leq M_{t_{j-1}}$. Assume moreover $\Phi_j(0) \leq 0$, and the nondegeneracy condition “ $M_{t_{j-1}} \geq 1$ or $A_j > 0$ ” holds. Then, the estimated clearing price equation (2) admits a unique solution.*

Proof. Clearly, Φ_j is continuous, $\lim_{p \rightarrow +\infty} \Phi_j(p) = +\infty$ and for $q > p$,

$$\Phi_j(q) - \Phi_j(p) = \sum_{i=1}^{M_{t_{j-1}}} [g_{i,t_{j-1}}(q) - g_{i,t_{j-1}}(p)] + A_j(q - p),$$

where $A_j \geq 0$. If $M_{t_{j-1}} \geq 1$, the first term is strictly positive, and if $M_{t_{j-1}} = 0$, then $A_j > 0$ and the second term is strictly positive. Hence Φ_j is strictly increasing. The intermediate value theorem gives existence of a zero, and strict monotonicity gives uniqueness. Since $\Phi_j(0) \leq 0$, the zero is nonnegative. $\square$

Theorem 2.1 provides conditions under which the clearing equation in our model admits a unique solution when the background participants' supply and demand schedules are not restricted to be linear. In practice, market participants submit collections of limit orders which the exchange aggregates to reconstruct the supply/demand functions. We refer to [46] for an example. These supply/demand functions need not be linear, and in fact that is not even standard in general. Finally and as we have mentioned earlier, our clearing rule recovers the one stated in [37] or [46] for linear supply and demand curve as stated in the following proposition.

Proposition 2.1. *Fix $j \in \{n+2, \dots, m+1\}$. Assume that $g_{i,t}(p) = K_t^i(p - S_t^i)$ with $K_t^i \geq 0$, $\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i + A_j > 0$ and $\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i S_{t_{j-1}}^i + B_j + C_j \geq 0$. Then there exists a unique clearing price solving (2) and given by*

$$p = \frac{\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i S_{t_{j-1}}^i + B_j + C_j}{\sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i + A_j} \geq 0.$$

For the linear specification used below, we consider background schedules satisfying $K_t^i > 0$ and impose the conditions of Proposition 2.1 at every auction evaluation:

$$D_j = \sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i + A_j > 0 \quad \text{and} \quad F_j = \sum_{i=1}^{M_{t_{j-1}}} K_{t_{j-1}}^i S_{t_{j-1}}^i + B_j + C_j \geq 0.$$

For each continuous root p_j^* in Equation (2), define its two bracketing ticks $p_j^- := \alpha \lfloor p_j^* / \alpha \rfloor$ and $p_j^+ := \alpha \lceil p_j^* / \alpha \rceil$, and let $\mathcal{P}_j := \{p_j^-, p_j^+\}$. Let $Q_j^S(p)$ and $Q_j^D(p)$ denote total supply and demand in the same book used to compute p_j^* , with the positive/negative-part definitions given below for the terminal book. Include market orders and aggregate all live strategic schedules into one net schedule before splitting its sign. Write $M_j(p) := \min\{Q_j^S(p), Q_j^D(p)\}$. The volume-maximizing projection, used for both lagged auction indications and terminal clearing, is

$$p_{\alpha,j} := \arg \max_{p \in \mathcal{P}_j} (M_j(p), -|p - p_j^*|, p),$$

where the tuple is ordered lexicographically: maximize matched volume first, then choose the tick nearest the root, then the higher tick. An on-grid root has one candidate. Monotone supply and demand ensure that a global tick-volume maximizer is among the two bracketing ticks. For linear books, nearest-root rounding minimizes absolute imbalance, whereas the volume-maximizing projection used here need not do so. The observation timing in Section 2.1.4 is unchanged. At the terminal evaluation, write $p_\alpha = p_{\alpha,m+1} = S_{\tau^{\text{cl}}}^{\text{cl}}$. The selected tick is at most one tick from the root, so for the linear aggregate

$$|Q_j^S(p_{\alpha,j}) - Q_j^D(p_{\alpha,j})| = D_j |p_{\alpha,j} - p_j^*| \leq D_j \alpha.$$

The agent's requested signed quantity at the selected clearing tick is

$$\tilde{Z}_{\tau^{\text{cl}}} = \sum_{s=n+1}^m \left(1 - \theta_{t_{m+1}}^{(s-n)}\right) K_{t_s}^a \left(S_{\tau^{\text{cl}}}^{\text{cl}} - S_{t_s}^a\right).$$

Let

$$\forall p \in \mathbb{R}_+, \quad \Sigma_{\tau^{\text{cl}}}^a(p) = \sum_{s=n+1}^m \left(1 - \theta_{t_{m+1}}^{(s-n)}\right) K_{t_s}^a (p - S_{t_s}^a).$$

At the selected terminal tick p_α , total supply and demand are

$$Q^S(p_\alpha) = \sum_{i=1}^{M_{t_m}} g_{i,t_m}(p_\alpha)_+ + \Sigma_{\tau^{\text{cl}}}^a(p_\alpha)_+ + \sum_{i=N_{\tau^{\text{op}}_-}^- + 1}^{N_{t_m}^-} \nu_{t_m}^{-,i},$$

and

$$Q^D(p_\alpha) = \sum_{i=1}^{M_{t_m}} (-g_{i,t_m}(p_\alpha))_+ + (-\Sigma_{\tau^{\text{cl}}}^a(p_\alpha))_+ + \sum_{i=N_{\tau^{\text{op}}_-}^+ + 1}^{N_{t_m}^+} \nu_{t_m}^{+,i}.$$

Let

$$\rho_S = \begin{cases} 1 & \text{if } Q^S = 0 \text{ or } Q^S \leq Q^D \\ Q^D/Q^S & \text{if } Q^S > Q^D \end{cases} \quad \text{and} \quad \rho_D = \begin{cases} 1, & \text{if } Q^D = 0 \text{ or } Q^D \leq Q^S \\ Q^S/Q^D & \text{if } Q^D > Q^S \end{cases}.$$

The actual executed volume of the agent at the clearing time is $Z_{\tau^{\text{cl}}} = \rho_S(\tilde{Z}_{\tau^{\text{cl}}})_+ - \rho_D(-\tilde{Z}_{\tau^{\text{cl}}})_+$. The agent's active schedules are aggregated into a single net schedule before this allocation, so no self-trade is counted. A negative terminal inventory is an admissible outcome of the enlarged auction action set rather than a violation of an inventory constraint. Deviations from zero are discouraged through the terminal inventory penalty.

2.1.4 Projected Hypothetical Clearing Price During the Continuous Session

During the continuous phase $[0, \tau^{\text{op}})$, the simulator uses the standing exogenous CLOB to construct a raw projected clearing estimator H_t° . It treats the outstanding exogenous orders as inputs to a fictitious auction and calibrates their linear-schedule representation through the volume moments in Algorithm 1, following the construction motivated by [37]. A separate training-only reliability fit, defined immediately after the algorithm, converts this raw estimator into the CLOB forecast H_t^{cl} supplied to the policy. The policy does not receive the complete order multiset or H_t° as an additional feature.

At t_i , the previous strategic CLOB order has already expired. Therefore, O_i denotes the standing exogenous CLOB multiset observed before the action at t_i .

Algorithm 1 Computation of the raw estimator $H_{t_i}^\circ$ for $i \in \llbracket 0, n \rrbracket$

Require: Tick size $\alpha > 0$, smoothing parameter $\eta_H \in (0, 1]$, initial value H_0

```

1: Initialize  $H_{t_0}^\circ \leftarrow H_0$ 
2: for  $i = 1, \dots, n$  do
3:   Record standing orders before time  $t_i$  by multiset  $O_i \in \mathcal{M}_f(\alpha\mathbb{N} \times \mathbb{R}_+)$   $\{\mathcal{M}_f(\cdot)$  denotes the
   finite multisets. $\}$ 
4:   if  $O_i \neq \emptyset$  then
5:      $\mathcal{K}_i \leftarrow \text{proj}_1(O_i)/\alpha$  {Standing price levels}
6:     Register  $Q_i(k) = \sum_{(p,v) \in O_i} v \delta_{p,\alpha k}$  for  $k \in \mathcal{K}_i$  and  $Q_i(k) = 0$  if  $k \notin \mathcal{K}_i$ 
7:     for  $k \in \mathcal{K}_i$  do
8:        $\hat{e}_i^k \leftarrow \frac{1}{i} \sum_{s=1}^i Q_s(k)$  {Average volume available at level  $k$ , with  $\hat{e}_i^k > 0$  for  $k \in \mathcal{K}_i$ }
9:        $\hat{\zeta}_i^k \leftarrow \frac{1}{i} \sum_{s=1}^i Q_s(k)^2$  {Average squared volume available at level  $k$ }
10:       $\hat{K}_i^k \leftarrow \max(0, (2\hat{e}_i^k - \hat{\zeta}_i^k/\hat{e}_i^k)\alpha^{-1})$  {Calibrated slope at level  $k$ }
11:    end for
12:    if  $\sum_{k \in \mathcal{K}_i} \hat{K}_i^k > 0$  then
13:      Solve  $\sum_{k \in \mathcal{K}_i} \hat{K}_i^k(\alpha k - p) = 0$  for  $p$  and denote the solution  $\tilde{S}_{t_i}$  {Clearing price rule}
14:       $H_{t_i}^\circ \leftarrow H_{t_{i-1}}^\circ + \eta_H(\tilde{S}_{t_i} - H_{t_{i-1}}^\circ)$  {Smoothed update rule}
15:    else
16:       $H_{t_i}^\circ \leftarrow H_{t_{i-1}}^\circ$ 
17:    end if
18:  else
19:     $Q_i(k) \leftarrow 0$  for every price level  $k$ 
20:     $H_{t_i}^\circ \leftarrow H_{t_{i-1}}^\circ$ 
21:  end if
22: end for
```

Before using the raw estimate as a forecast, we shrink its displacement from the current midprice:

$$H_t^{\text{cl}} = S_t^{\text{mid}} + w_{b(t)}(H_t^\circ - S_t^{\text{mid}}).$$

Here $b(t) \in \llbracket 0, l \rrbracket$ identifies the time bin containing t . We consider $l = 3$ from now on. For example, when working on a time horizon of 120 minutes, the four equal quarters before opening are $[0, 30)$, $[30, 60)$, $[60, 90)$ and $[90, 120)$ minutes in the numerical setting. This correction neither feeds back into Algorithm 1 nor rounds the CLOB forecast to a tick. Initially, the raw estimate and midprice both equal 100. We fit the weights separately on 256 no-order training paths for each setting. Within each bin, we regress the eventual clearing price's displacement from the

current midprice on the corresponding raw-estimate displacement. The regression has no intercept and constrains the coefficient to $[0, 1]$; a bin with zero predictor sum of squares receives weight zero. Historical calibration draws only from the pooled training sessions, and all order flow and clearing on these paths are simulated. The fitted weights for the four successive quarters are $(0.075, 0.112, 0.225, 0.340)$ on synthetic paths and $(0.192, 0.312, 0.384, 0.523)$ on historical paths. Thus the fit puts more weight on the raw estimate as closing approaches. The weights are then frozen across learners and treatments. No validation or test outcome enters the fit. This forecast calibration is separate from the moment-based slope calculation and the residual-book conversion below.

At auction opening, the environment inherits the last calibrated pre-action CLOB forecast:

$$H_{\tau^{\text{op}}}^{\text{cl}} = H_{t_n}^{\text{cl}} = S_{t_n}^{\text{mid}} + w_{b(t_n)}(H_{t_n}^{\circ} - S_{t_n}^{\text{mid}}).$$

It is not replaced by the residual-book raw estimate or recalibrated around the newly frozen opening midprice. For $j = n + 2, \dots, m$, let p_j^* be the continuous root of (2), computed from the complete post-action book at t_{j-1} , including all strategic schedules still live after that action. The next decision's indicative price is $H_{t_j}^{\text{cl}} = p_{\alpha, j}$. The indicative price uses the same volume-maximizing projection as terminal settlement, applied to this lagged book. It is an indication of hypothetical clearing, with no execution before τ^{cl} . Current exogenous proposals at t_j are accepted before the agent observes the current features and acts, but do not update this lagged indicative price. No CLOB reliability weight is applied during the auction. The final post-action book at t_m is frozen; at $t_{m+1} = \tau^{\text{cl}}$, $H_{\tau^{\text{cl}}}^{\text{cl}} = S_{\tau^{\text{cl}}}^{\text{cl}}$ is its volume-maximizing terminal clearing price defined in Section 2.1.3. The clearing conditions and actual pro-rata allocation of Section 2.1.3 are unchanged.

In the preference objective defined next, the CLOB multiplier compares the execution price with the calibrated H_t^{cl} , not with H_t° . This is a specified preference for execution relative to anticipated clearing, not a guarantee that waiting for the auction improves economic performance.

Let $O_{\tau^{\text{op}}-}^*$ denote the multiset of exogenous CLOB orders remaining after all executions on $(t_n, \tau^{\text{op}}]$. The unexecuted remainder of the agent's order submitted at t_n expires at τ^{op} and is excluded from this multiset. To map the residual CLOB book into the linear-schedule representation used in the auction, we repeat the final calibration step of Algorithm 1 with the residual multiset $O_{\tau^{\text{op}}-}^*$ in place of the pre-action multiset O_n , while the earlier snapshots $O_1, \dots, O_{n-1}$ are left unchanged. Denote the resulting slopes by $\hat{K}_{\star}^k$ and let

$$\mathcal{K}_{\star} = \left\{ k : O_{\tau^{\text{op}}-}^* \text{ contains positive volume at } \alpha k \text{ and } \hat{K}_{\star}^k > 0 \right\}.$$

At the auction opening, we assume that the initial exogenous auction book contains the carry-over schedules $g_{k, \tau^{\text{op}}}^*(p) = \hat{K}_{\star}^k(p - \alpha k)$ for $k \in \mathcal{K}_{\star}$. These schedules are included in $M_{\tau^{\text{op}}}$ before any auction-specific arrivals are processed. Let $D^* = \sum_{k \in \mathcal{K}_{\star}} \hat{K}_{\star}^k$ be the aggregate carry-over slope. If $D^* < D_{\mu}$, for some numerical lower slope bound $D_{\mu} > 0$, we add the “persistent” fallback schedule $g^0(p) = (D_{\mu} - D^*)(p - S_{\tau^{\text{op}}}^{\text{mid}})$ which cannot be canceled. If $D^* \geq D_{\mu}$, one carry-over schedule with maximal slope is designated as “persistent”.

Remark 2.6. *This construction is a stylized linear representation of the residual exogenous CLOB book. It does not preserve the original side of each individual CLOB order. Instead, each retained price level becomes the zero of a two-sided linear net-supply schedule centered at that level.*

3 Learning Problem for Optimal Liquidation with Closing Auction

We now turn to the discretization of the problem.

Definition 3.1. Set $t_0 = 0$. For every $i \geq 0$ such that $t_i < \tau^{\text{op}} - 1$ and every $\zeta \in \{+, -\}$, define $\tau_i^\zeta = \inf\{s > t_i : N_s^\zeta > N_{t_i}^\zeta\}$ and $\tau_i := \tau_i^+ \vee \tau_i^-$ with the convention $\inf \emptyset = +\infty$. We set $t_{i+1} = (\tau^{\text{op}} - 1) \wedge [\tau_i \vee (1 + \lfloor t_i \rfloor)]$ until the bound $\tau^{\text{op}} - 1$ is attained. The final CLOB decision index is $n = \inf\{i \geq 0 : t_i = \tau^{\text{op}} - 1\}$. Thus there are $n + 1$ CLOB decisions, $t_0, \dots, t_n$, and we set $t_{n+1} = \tau^{\text{op}}$ and $t_{n+1+\ell} = \tau^{\text{op}} + \ell$ for $\ell \in \llbracket 0, h \rrbracket$. Finally, $m = n + h$, so that $t_{m+1} = \tau^{\text{cl}}$, and $\mathcal{T} = (t_i)_{0 \leq i \leq m+1}$.

Remark 3.1. Except for the final capped interval ending at $\tau^{\text{op}} - 1$, the maximum $\tau_i^+ \vee \tau_i^-$ means that a new decision occurs only after both sides have received at least one additional market order. For simulation, the realized grid is a functional of the exogenous order-flow path. At decision i , however, only $t_0, \dots, t_i$ and information available at t_i are revealed to the agent. The last CLOB decision occurs at the deterministic physical time $t_n = \tau^{\text{op}} - 1$, while the random index n determines how many earlier CLOB decisions occurred.

In the following, i denotes the decision index whereas t_i denotes the corresponding physical time. We do not identify these two quantities during the continuous phase, since $(t_i)_{0 \leq i \leq n}$ is random. Note however that during the auction phase the two are related explicitly by $t_{n+1+\ell} = \tau^{\text{op}} + \ell$ for $\ell \in \llbracket 0, h \rrbracket$. For simplicity, albeit an abuse of notation, when there is no ambiguity, we write a generic decision time as t .

The latent state is denoted by X and defined for any time $t \in \mathcal{T}$ as a tuple

$$X_t = (t, X_t^1, X_t^2, X_t^3, X_t^4, X_t^5, X_t^6, X_t^7, X_t^8, X_t^9, X_t^{10}, X_t^{11}, X_t^{12}, X_t^{13}, X_t^{14}, X_t^{15}, X_t^{16}, X_t^{17}, X_t^{18}),$$

where each attribute encodes one of the market characteristics before the choice of an action by the agent as detailed below.

State Variables

- Inventory: $X_t^1 = I_t$ represents the inventory of the market maker at time t : per construction, we restrict $I_t \geq 0$ for $t < \tau^{\text{op}}$ and I_t frozen for $\tau^{\text{op}} \leq t < \tau^{\text{cl}}$ with $I_{\tau^{\text{cl}}} \in \mathbb{R}$;
- Volume executed at the clearing: $X_t^2 = 0$ for $t < \tau^{\text{cl}}$ with $X_{\tau^{\text{cl}}}^2 = Z_{\tau^{\text{cl}}}$ and $H_{\tau^{\text{cl}}}^{\text{cl}} = S_{\tau^{\text{cl}}}^{\text{cl}}$;
- Calibrated forecast/indicative-price history: $X_t^3 = (H_s^{\text{cl}})_{\mathcal{T} \ni s \leq t}$ records the calibrated CLOB forecasts of Section 2.1.4 for $s < \tau^{\text{op}}$, the inherited calibrated forecast $H_{\tau^{\text{op}}}^{\text{cl}} = H_{t_n}^{\text{cl}}$ at opening, and the subsequent lagged, tick-projected indicative prices from (2). The raw estimator H_s^{o} is internal to Algorithm 1, not this observed forecast history;
- Limit order book depth: $X_t^4 = L_t^+ \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$ and $X_t^5 = L_t^- \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$ represent respectively the depth in the limit order book on the ask (resp. bid) side;
- Number of limit orders in the auction: $X_t^6 = M_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}$;
- Number of accepted market-order arrivals in the auction: $X_t^7 = X_t^8 = 0$ for $t < \tau^{\text{op}}$, while $X_t^7 = N_t^+ - N_{\tau^{\text{op}}-}^+$ and $X_t^8 = N_t^- - N_{\tau^{\text{op}}-}^-$ for $t \geq \tau^{\text{op}}$ count buy and sell arrivals, respectively, including subsequently canceled orders;
- Cancellation history: $X_t^9 = \theta_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}$ represents the vector of canceled orders in the auction up to time t ;
- Mid price: $X_t^{10} = S_t^{\text{mid}}$

- Auction market-order volumes on the buy and sell sides, respectively: $X_t^{11} = X_t^{12} = \emptyset$ for $t < \tau^{\text{op}}$, and for $t \geq \tau^{\text{op}}$, $X_t^{11} = (\nu_t^{+,i})_{N_{\tau^{\text{op}}_-}^+ < i \leq N_t^+}$, and $X_t^{12} = (\nu_t^{-,i})_{N_{\tau^{\text{op}}_-}^- < i \leq N_t^-}$.
- Volume in the limit order book (ask/bid side): $X_t^{13} = (V_t^{+,j} \mathbf{1}_{\{t \leq \tau^{\text{op}} - 1\}})_{1 \leq j \leq L_t^+}$ and $X_t^{14} = (V_t^{-,j} \mathbf{1}_{\{t \leq \tau^{\text{op}} - 1\}})_{1 \leq j \leq L_t^-}$ are the volume existing in the limit order book on the ask and bid side at any depth
- Limit order in the auction: $X_t^{15} = (g_{i,t})_{1 \leq i \leq M_t}$, with $X_t^{15} = \emptyset$ before the auction;
- Price history in the auction: $X_t^{16} = S^a(t)$ where $S^a(t) := (S_{t_{n+1}}^a, \dots, S_{t-1}^a, 0, \dots, 0)$ represents the vector of limit order prices submitted in the auction up to time t ;
- Supply/Demand slope history: $X_t^{17} = K^a(t)$ where $K^a(t) := (K_{t_{n+1}}^a, \dots, K_{t-1}^a, 0, \dots, 0)$ represents the slope of the limit order submitted up to time t ;
- Decision index: $X_t^{18} = i$ if $t = t_i$ is the time of the i -th decision: let us denote i_t the decision index of time t for a generic time $t \in \mathcal{T}$.

Remark 3.2. In the linear specification of Proposition 2.1 and in the numerical simulator, $g_{i,t}(p) = K_t^i(p - S_t^i)$ so X_t^{15} can equivalently be represented by $((K_t^i, S_t^i))_{1 \leq i \leq M_t}$.

Remark 3.3. Note that X^{16}, X^{17} are vectors of size $h = m - n$ with 0 components after time t .

The state vector X_t is viewed as an element of the state space $\mathcal{X}$ induced by the above variables, that is

$$\mathcal{X} = [0, \tau^{\text{cl}}] \times \mathbb{R}^2 \times \Psi(\mathbb{R}_+) \times \mathbb{N}^5 \times \{0, 1\}^h \times \alpha\mathbb{N} \times \Psi(\mathbb{R}_+)^4 \times \Psi(\mathcal{G}) \times (\alpha\mathbb{N})^h \times \mathbb{R}_+^h \times \mathbb{N}$$

where

$$\mathcal{G} = \left\{ g \in \mathcal{C}(\mathbb{R}_+) : g \text{ is strictly increasing, } \lim_{p \rightarrow +\infty} g(p) = +\infty \right\}$$

is the space of functions that satisfy the assumptions of Theorem 2.1, and for a set E , the space $\Psi(E)$ is defined as

$$\Psi(E) = \bigsqcup_{k=0}^{+\infty} E^k$$

with $E^0 = \{\emptyset\}$.

Remark 3.4. While exchanges like the NYSE usually only display statistics on the total imbalance between buy and sell orders combined with an indicative price, they do not display all the orders that have been submitted at each price.

Theorem 2.1 permits general exogenous schedules. From this point onward, the learning problem and the numerical simulator use the linear specialization of Proposition 2.1 and Remark 3.2. Accordingly, the aggregate slope and intercept features used below are well defined. The tuple X_t records the market configuration used in this formulation, while the learning algorithms receive only the reduced feature vector $\varphi(X_t)$. The simulator additionally retains internal memory needed to generate future prices and update the projected-clearing estimator, including the rough-Heston history or selected historical session and the running volume moments of Algorithm 1. Neither the

displayed configuration nor its feature vector is asserted to be Markov sufficient. This reduced state is $\varphi(X_t)$ with

$$\begin{aligned}\varphi(X_t) = & (t, I_t, H_t^{\text{cl}}, S_t^{\text{mid}}, i_t, L_t^+ \mathbf{1}_{\{t < \tau^{\text{op}}\}}, L_t^- \mathbf{1}_{\{t < \tau^{\text{op}}\}}, V_t^{+,1} \mathbf{1}_{\{t < \tau^{\text{op}}\}}, V_t^{-,1} \mathbf{1}_{\{t < \tau^{\text{op}}\}}, \\ & M_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, X_t^7, X_t^8, C(X_t) \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, A_{it} \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, B_{it} \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, \\ & G_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, \mathcal{I}_t \mathbf{1}_{\{t \geq \tau^{\text{op}}\}}, R_t^e \mathbf{1}_{\{t \geq \tau^{\text{op}}\}})\end{aligned}$$

with A_{it} and B_{it} given by the expressions of A_j and B_j in Section 2.1.3, $C(x) = \max_{1 \leq i \leq m-n} \{(1 - x_i^9) \mathbf{1}_{\{x_i^{17} > 0\}}\}$ for a state $x \in \mathcal{X}$, and

$$G_t = \sum_{i=1}^{M_t} K_t^i, \quad R_t^e = \sum_{i=1}^{M_t} K_t^i S_t^i, \quad \mathcal{I}_t = \sum_{i=N_{\tau^{\text{op}}-}^+ + 1}^{N_t^+} \nu_t^{+,i} - \sum_{i=N_{\tau^{\text{op}}-}^- + 1}^{N_t^-} \nu_t^{-,i}.$$

Let $\mathcal{Y} = \varphi(\mathcal{X})$.

The displayed feature vector is the raw H-on observation: its third coordinate is the calibrated CLOB forecast or the phase-appropriate auction indicative price just defined, not H_t° . In H-off treatments, the forecast coordinate is set to zero after the network-input transformation; the internal forecast computation, raw moment recursion and residual-book conversion are retained. Other clearing-relevant observations, including midprice, CLOB liquidity and auction aggregates, remain available. Removing the H feature is therefore not equivalent to removing all clearing information, and does not establish Markov sufficiency of the remaining features.

Action Variables We now turn to the actions of the strategic market maker. Given a state vector $X_t \in \mathcal{X}$, we define the action vector A_t as $A_t = (A_t^1, A_t^2, A_t^3, A_t^4, A_t^5)$ where each component represents a particular action.

- Volume set in the limit order: $A_t^1 = v_t \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$;
- Depth offset in the limit order book: $A_t^2 = \delta_t \mathbf{1}_{\{0 \leq t \leq \tau^{\text{op}} - 1\}}$;
- Supply/demand slope and reference price: $A_t^3 = K_t^a \mathbf{1}_{\{\tau^{\text{op}} \leq t \leq \tau^{\text{cl}} - 1\}}$; and $A_t^4 = \ell_t^a \mathbf{1}_{\{\tau^{\text{op}} \leq t \leq \tau^{\text{cl}} - 1\}}$ such that $S_t^a = S_{\tau^{\text{op}}}^{\text{mid}} + \alpha b_t^a$ with

$$b_t^a = (b^{\text{anc}}(\varphi(X_t)) + \ell_t^a) \mathbf{1}_{\{K_t^a > 0\}}$$

- Order cancellation in the auction: $A_t^5 = c_t \mathbf{1}_{\{\tau^{\text{op}} \leq t \leq \tau^{\text{cl}} - 1\}}$.

For the selected anchor convention, define the integer anchor offset for a raw observation y by

$$b^{\text{anc}}(y) = \begin{cases} \left\lfloor \frac{y_3 - y_4}{\alpha} + \frac{1}{2} \right\rfloor, & \text{indicative anchor (headline),} \\ 0, & \text{frozen-mid anchor.} \end{cases}$$

During the auction, $y_4 = S_{\tau^{\text{op}}}^{\text{mid}}$ and $y_3 = H_t^{\text{cl}}$, including the inherited calibrated CLOB forecast at opening. Thus the headline retains indicative anchoring. Under the frozen-mid anchor, a positive-slope schedule instead has $b_t^a = \ell_t^a$ and $S_t^a = S_{\tau^{\text{op}}}^{\text{mid}} + \alpha \ell_t^a$. Zero slope uses the canonical $\ell_t^a = b_t^a = 0$; its reference price has no effect on the book. The same anchor offset is used in the action mapping and admissibility conditions below, including the absolute price-deviation and nonnegative-price

constraints. Changing the anchor shifts the executable price coverage of the bounded local offset grid; it does not change the forecast definition or remove the H feature.

Anchoring and H-feature availability are separate treatment switches. The matched H-on/H-off comparison uses a frozen-mid anchor and economic-only rewards in both arms, so the explicit H signal is not supplied through a forecast-dependent anchor or preference reward. H-off still retains the other clearing-relevant observations described above. DQN and the projected continuous-proposal methods use the same selected anchor convention in their executable controls.

We view such actions as part of the ambient action set

$$\mathcal{A} = \llbracket 0, V_{\max} \rrbracket \times \llbracket 0, L_{\max} \rrbracket \times \beta \llbracket 0, K_{\max} \rrbracket \times \llbracket -B_{\max}, B_{\max} \rrbracket \times \{0, 1\}$$

where $V_{\max}, L_{\max}, B_{\max}, K_{\max} \in \mathbb{N}^*$ are numerical bounds defining the finite action grid.

Recalling that on the limit order book the market maker is liquidating his inventory, hence submits a volume $A_t^1 \leq I_t = X_t^1$, at a price $S_t^{\text{mid}} + \alpha A_t^2 \geq S_t^{\text{mid}}$. At an auction decision, $c_t = 1$ is admissible if and only if at least one prior positive-slope strategic schedule is live. It cancels every such schedule submitted strictly before t and the current schedule is inserted afterward. As $\theta_t^{(s)} = 1$ if and only if the positive-slope schedule submitted in slot s has already been canceled (with $\theta_t^{(s)} = 0$ if that slot contained no order), we obtain the admissibility condition $A_t^5 = c_t \leq \max_{1 \leq i \leq t - \tau^{\text{op}}} (1 - \theta_t^{(i)}) \mathbf{1}_{\{K_{t_n+i}^a > 0\}} =: C_t$ with the convention that $\max \emptyset = 0$ (for $t \leq \tau^{\text{op}}$).

Definition 3.2 (Admissible actions). *Given an observed state $y \in \mathcal{Y}$, the set $\text{Adm}(y)$ of admissible actions is defined as*

$$\begin{aligned} \text{Adm}(y) = \{a \in \mathcal{A} : & a_1 \leq y_2 \mathbf{1}_{\{y_1 < \tau^{\text{op}}\}}, a_5 \leq y_{13}, y_4 + \alpha(b^{\text{anc}}(y) + a_4) \mathbf{1}_{\{a_3 > 0\}} \geq 0, \\ & a_1 = 0 \implies a_2 = 0, a_3 = 0 \implies a_4 = 0, (a_1, a_2) \mathbf{1}_{\{y_1 \geq \tau^{\text{op}}\}} = 0, \\ & (a_3, a_4, a_5) \mathbf{1}_{\{y_1 < \tau^{\text{op}}\}} = 0, |a_4 + b^{\text{anc}}(y)| \mathbf{1}_{\{a_3 > 0\}} \leq B_{\infty}\} \end{aligned}$$

where $B_{\infty} > 0$ and the last two components embed the phase compatibility of admissible actions.

Remark 3.5. *Note that S_t^{mid} is constant during the auction. Furthermore, since all X_t^1 , X_t^{10} and $C(X_t)$ are encoded in $\varphi(X_t)$, all admissible actions for X_t are exactly those admissible for $\varphi(X_t)$.*

Definition 3.3 (Admissible policies). *An admissible policy is a measurable map*

$$\pi : y \in \mathcal{Y} \longmapsto \pi(\cdot | y) \in \mathcal{P}(\mathcal{A})$$

such that $\pi(\text{Adm}(y) | y) = 1$ for all $y \in \mathcal{Y}$. We denote Π the set of these admissible policies.

Reward The weighted preference objective J uses the rewards below, before initial-value centering and learning-only conditioning. The fictive auction-credit weight is $\omega_A \geq 0$, with headline value 10^{-4} . It weights interim credit and terminal purchase attenuation, not actual cash, fees, residual marking or inventory risk. The CLOB multiplier f^c is a separate preference. The rewards use the calibrated or lagged H_t^{cl} defined in Section 2.1.4 and being the last component of X_t^3 , not H_t° . The auction anchor changes S_t^a , not H_t^{cl} .

1. During the continuous trading session for $t < \tau^{\text{op}}$. The market maker submits a price $S_t^{\bullet} = S_t^{\text{mid}} + \alpha A_t^2$. The volume executed is given by E_t . The cash proceeds are thus given by $S_t^{\bullet} \times E_t$. We moreover assume that the market maker penalizes the execution by comparing the price executed with the hypothetical clearing price $H_t^{\text{cl}} = \varphi(X_t)_3$. If $S_t^{\bullet} > H_t^{\text{cl}}$, the cash proceeds

enter the reward without attenuation; otherwise if $S_t^\bullet < H_t^{\text{cl}}$ the market maker may regret the execution. The tolerated shortfall of the execution price $S_t^\bullet$ below the current scalar forecast H_t^{cl} is $k^\star \alpha$, for fixed $k^\star$. In the numerical specification, the CLOB reward multiplier is strictly positive when

$$(H_t^{\text{cl}} - S_t^\bullet)_+ < k^\star \alpha,$$

and vanishes when this gap is at least $k^\star \alpha$. It equals one when $S_t^\bullet \geq H_t^{\text{cl}}$. We thus introduce a penalty function $f^c : \mathbb{R} \rightarrow [0, 1]$ continuous and increasing such that f^c is zero on $\mathbb{R}_-$ and $f^c(k^\star \alpha) = 1$, such that the reward of the market maker is given by

$$r_t(X_t, A_t) = S_t^\bullet E_t f^c(k^\star \alpha - (H_t^{\text{cl}} - S_t^\bullet)_+).$$

2. During the auction trading session for $\tau^{\text{op}} \leq t < \tau^{\text{cl}}$. The market maker submits a slope $K_t^a = A_t^3$ and a price

$$S_t^a = S_{\tau^{\text{op}}}^{\text{mid}} + \alpha b_t^a, \quad b_t^a = (b^{\text{anc}}(\varphi(X_t)) + A_t^4) \mathbf{1}_{\{K_t^a > 0\}}.$$

The projected cash $K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a)$ is not settlement cash. Purchase attenuation uses a convex, continuous, nonincreasing function $f^a : \mathbb{R} \rightarrow \mathbb{R}_+$, with $f^a(x) = 0$ for $x \geq 0$ and $f^a(x) \leq (-x)_+$. The weighted schedule credit and auction reward are

$$u_t = \omega_A \left[K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a) + f^a \left(K_t^a H_t^{\text{cl}} (H_t^{\text{cl}} - S_t^a) \right) \right],$$

and

$$r_{t_j}(X_{t_j}, A_{t_j}) = u_{t_j} - c_{t_j} \left[d_{t_j} + \sum_{i=n+1}^{j-1} \mathbf{1}_{\{K_{t_i}^a > 0\}} (1 - \theta_{t_j}^{(i-n)}) u_{t_i} \right].$$

The cancellation charge is $d_{t_j} c_{t_j}$, with a rate determined by elapsed auction time. The fictive schedule credits are noncash preference terms in J , distinct from the policy-invariant learning conditioning of Section 4.1.

3. Final reward at the clearing for $t = \tau^{\text{cl}}$. At the clearing time, the clearing price $S_{\tau^{\text{cl}}}^{\text{cl}}$ is determined and orders are matched. The signed execution cash is $S_{\tau^{\text{cl}}}^{\text{cl}} Z_{\tau^{\text{cl}}}$ for the actual allocation of her live schedules. The agent is furthermore penalized for holding inventory. We introduce $\lambda > 0$ as a penalization parameter. Furthermore, the agent's reward associated with purchases is once again attenuated. Since the midprice is frozen throughout the auction phase in our model, any inventory remaining after the auction is priced at the last exogenous midprice $S_{\tau^{\text{op}}}^{\text{mid}}$ rather than at the auction clearing price. This separates the valuation of residual inventory from the endogenous auction price, which may be affected by the agent's own submitted schedules. The final reward of the market maker is given by

$$G(X_{\tau^{\text{cl}}}) = S_{\tau^{\text{cl}}}^{\text{cl}} Z_{\tau^{\text{cl}}} - \lambda |I_{\tau^{\text{cl}}}|^2 + \omega_A f^a \left(S_{\tau^{\text{cl}}}^{\text{cl}} Z_{\tau^{\text{cl}}} \right) + S_{\tau^{\text{op}}}^{\text{mid}} I_{\tau^{\text{cl}}}.$$

Terminal attenuation acts once on actual aggregate pro-rata cash, not on individual schedules. The allocation is defined in Section 2.1.3; terminal inventory may have either sign.

Remark 3.6. Numerically, $f^c(t) = \min\{1, t_+/(k^\star \alpha)\}$ and $f^a(t) = q(-t)_+$ for $q \in [0, 1]$. The headline has $q = 0$: purchase attenuation disappears, but the CLOB multiplier, weighted interim credit and exact cancellation clawback remain active.

In our setting, the agent chooses action A_{t_i} in state X_{t_i} . Then, the executed volume E_{t_i} is realized, determining the one-step reward $r_{t_i}(X_{t_i}, A_{t_i})$ during the state transition. Finally, the agent transitions into state $X_{t_{i+1}}$.

The headline training objective is the weighted manuscript preference objective J , defined by the rewards r_t and G above:

$$\begin{aligned}
(\mathbf{P}) \quad & \sup_{\pi \in \Pi} J(\pi) = \mathbb{E}_{\pi} \left[\sum_{i=0}^m r_{t_i}(X_{t_i}, A_{t_i}) + G(X_{\tau\text{cl}}) \right] \\
\text{s.t.} \quad & \begin{cases} X_0 \sim \mu_0 \\ A_{t_i} \sim \pi(\cdot \mid \varphi(X_{t_i})) \end{cases} .
\end{aligned}$$

3.1 Implementation Shortfall

The actual signed auction fill is $Z_{\tau\text{cl}}$, its clearing price is $S_{\tau\text{cl}}^{\text{cl}}$, and residual inventory is $I_{\tau\text{cl}} = I_{\tau\text{op}} - Z_{\tau\text{cl}}$. Relative to immediate execution at the initial midprice, episode implementation shortfall is

$$\text{IS} = S_0^{\text{mid}} I_0 - \left[\sum_{i=0}^n S_{t_i}^{\bullet} E_{t_i} + S_{\tau\text{cl}}^{\text{cl}} Z_{\tau\text{cl}} + S_{\tau\text{op}}^{\text{mid}} I_{\tau\text{cl}} - \sum_{i=n+1}^m d_{t_i} c_{t_i} \right]. \quad (3)$$

Positive shortfall denotes underperformance of the initial-midprice benchmark; negative shortfall denotes improvement, and lower is better. It is the negative of the marked-to-market economic PnL defined by the same cash flows. The benchmark is an accounting reference, not an additional feasible execution in the simulator. Residual inventory is valued at the frozen auction-opening midprice, not the agent-influenced clearing price. This mark is not an executed trade. Auction fills and residual positions may have either sign, as permitted by Section 2.1.3. Shortfall includes modeled withdrawal fees and execution prices; no additional commissions, taxes or unmodeled costs are imputed. The separate inventory preference gives

$$\text{IS}_{\lambda} = \text{IS} + \lambda I_{\tau\text{cl}}^2. \quad (4)$$

The quadratic term is neither an actual cash expense nor part of ordinary implementation shortfall. Minimizing $\mathbb{E}_{\pi}[\text{IS}_{\lambda}]$ is equivalent to maximizing the former risk-adjusted criterion: $\bar{J}(\pi) = -\mathbb{E}_{\pi}[\text{IS}]$ and $\bar{J}_{\lambda}(\pi) = -\mathbb{E}_{\pi}[\text{IS}_{\lambda}]$.

Economic-only training replaces the CLOB multiplier by one and omits interim credits, their clawbacks and terminal purchase attenuation, while retaining actual cash, fees, residual marking and terminal inventory risk. These preferences change J ; initial-value centering and learning-only conditioning are treated separately in Section 4.1. Neither shortfall measure includes preferences or conditioning adjustments. Headline training uses weighted J , while validation checkpoint selection and policy comparisons minimize expected inventory-penalized shortfall, and SAC additionally uses phase-specific, automatically tuned proposal-entropy regularization.

4 Learning Algorithms

4.1 Numerical Representation and Learning Conditioning

The policy receives the common 18-coordinate raw observation $y_t = \varphi(X_t)$, not the full simulator configuration. Network inputs $\tilde{y}_t = \mathcal{N}(y_t)$ replace absolute forecast prices by $H_t^{\text{cl}} - S_t^{\text{mid}}$ and the

two weighted-quote coordinates by continuous-root residual exposure and displacement. Define

$$\begin{aligned} W_t^a &= B_{i_t} - S_t^{\text{mid}} A_{i_t}, & W_t^e &= R_t^e - S_t^{\text{mid}} G_t, \\ \Delta_t &= \frac{W_t^a + W_t^e + \mathcal{I}_t}{A_{i_t} + G_t}, & \hat{Z}_t &= A_{i_t} \Delta_t - W_t^a. \end{aligned}$$

For positive aggregate slope, the replacement coordinates are $I_t - \hat{Z}_t$ and Δ_t ; otherwise they are zero. This continuous-root exposure does not substitute for actual pro-rata settlement. DQN, DDPG and TD3 use phase-specific normalization and signed-asinh inventory transforms; SAC uses pooled linear normalization. Algorithm comparisons include these representation differences.

DQN, DDPG and TD3 apply $x \mapsto \text{asinh}(x/I_s)$, with $I_s = \alpha/\lambda$, to auction inventory and continuous-root residual inventory when aggregate slope is positive. CLOB inventory remains linear; SAC omits this nonlinear transform.

Normalization uses 16 dedicated training trajectories, cycling four equally represented sampling modes: random admissible CLOB actions followed by auction actions without cancellation; random actions in both phases; CLOB abstention followed by auction actions without cancellation; and random CLOB actions followed by auction abstention. Actions are uniform within each mode's admissible set. Only pre-action observations enter the fit. DQN/DDPG/TD3 fit separate population statistics for the CLOB and auction; SAC pools both phases. Time and decision index are divided by τ^{cl} , and the cancellation indicator is left unchanged. All other transformed coordinates are standardized using the fitted mean and population standard deviation; a standard deviation at most 10^{-12} is replaced by one.

Auction statistics apply at or after opening when aggregate slope is positive; otherwise CLOB statistics apply. H-off sets the forecast input to zero. Statistics are frozen before learning and retained with the selected policy.

The reference inventory curve averages the 12 calibration paths outside the CLOB-abstention mode. Each inventory history, including opening inventory, is linearly interpolated to the one-minute grid; the resulting mean curve is linearly interpolated at decision-interval midpoints. Historical fitting uses only the pooled training dates. Forecast fitting uses the separate no-order samples described in Section 2.1.4.

Write $y_i = \varphi(X_{t_i})$ at decision indices. Let $R_i^0 = r_{t_i}(X_{t_i}, A_{t_i}) + \mathbf{1}_{\{i=m\}} G(X_{\tau^{\text{cl}}})$. Initial-value centering yields

$$R_i = R_i^0 - S_0^{\text{mid}} [\mathbf{1}_{\{i \leq n\}} E_{t_i} + \mathbf{1}_{\{i=m\}} (Z_{\tau^{\text{cl}}} + I_{\tau^{\text{cl}}})].$$

Inventory conservation makes the total subtraction $S_0^{\text{mid}} I_0$, independent of policy. The reward used for learning evaluates the reference inventory at each interval midpoint:

$$\tilde{R}_i = R_i + \Phi(y_{i+1}) - \Phi(y_i) - \mathbf{1}_{\{i \leq n\}} q_{\text{ref}} \left(\frac{t_i + t_{i+1}}{2} \right) (S_{t_{i+1}}^{\text{mid}} - S_{t_i}^{\text{mid}}), \quad \Phi(y_{m+1}) = 0.$$

The last CLOB interval is included in the reference adjustment. The potentials are evaluated before input normalization:

$$\Phi(y_t) = \begin{cases} (S_t^{\text{mid}} - S_0^{\text{mid}}) I_t - \lambda [I_t - I_0(1 - t/\tau^{\text{op}})]_+^2, & t < \tau^{\text{op}}, \\ (S_{\tau^{\text{op}}}^{\text{mid}} - S_0^{\text{mid}}) I_t + \Delta_t \hat{Z}_t - \lambda (I_t - \hat{Z}_t)^2, & \tau^{\text{op}} \leq t < \tau^{\text{cl}}. \end{cases}$$

The CLOB depletion corridor is a potential baseline, not an inventory constraint. Sparse auction credit omits $\Delta_t \hat{Z}_t - \lambda (I_t - \hat{Z}_t)^2$, leaving a constant inventory-mark potential between opening and

settlement. All schemes retain actual terminal marking and risk; their enabled conditioning terms are listed in Table 2.

For either auction potential, summation gives

$$\sum_{i=0}^m \tilde{R}_i = \sum_{i=0}^m R_i - \Phi(y_0) - \sum_{i=0}^n q_{\text{ref}} \left(\frac{t_i + t_{i+1}}{2} \right) (S_{t_{i+1}}^{\text{mid}} - S_{t_i}^{\text{mid}}).$$

Here $\Phi(y_0) = 0$. Conditional on a fixed fitted reference curve, the last sum is policy-independent on common exogenous midprice paths and control-independent decision times. With a common initial state and integrable rewards, these adjustments preserve policy differences pathwise and expected rankings. They need not have zero expectation under historical prices, nor do they imply Markov sufficiency. Different fitted reference curves need not produce comparable learning-return levels.

In the headline, $\mathbb{E}_\pi[\sum_i R_i] = J(\pi) - S_0^{\text{mid}} I_0$. Under centered economic training, $\sum_i R_i = -\text{IS}_\lambda$; without centering, the economic reward sum is $S_0^{\text{mid}} I_0 - \text{IS}_\lambda$. Terminal settlement appears once in each identity. Potential and reference adjustments do not enter J or either shortfall measure.

For episode index e , structured exploration uses CLOB mode $\lfloor e/4 \rfloor \bmod 4$ and auction mode $e \bmod 4$. CLOB modes are maximum volume at offset $\min(1, L_{\text{max}})$, abstention, and two modes with episode-fixed uniformly sampled integer volume and offset. Inventory caps and zero-volume canonicalization still apply. Auction modes are abstention, nonpositive-offset schedules, nonnegative-offset schedules, and nonpositive-offset schedules with cancellation. Slope indices and offset magnitudes are drawn uniformly from their grids once per episode; zero slope uses zero offset. An auction target $(\bar{k}, \bar{\ell}, \bar{c})$ is mapped to the first admissible grid point minimizing

$$|K^a/\beta - \bar{k}| + |\ell^a - \bar{\ell}| + 100|c - \bar{c}|.$$

All learners use this design for the exploration period in Table 1. Subsequently DQN uses the global-episode linear epsilon schedule in that table, clipped to its start and end values before and after the stated interval; the episode clock is not restarted. Conditional on exploration, CLOB actions are uniform over the admissible set. Auction exploration assigns the listed mixture probability to uniform zero-slope controls and the remainder to uniform sampling over all admissible actions. Greedy ties use fixed grid order. Continuous learners with insufficient phase-buffer observations still use uniform proposals until their threshold is met.

4.2 Deep Q-Network

DQN uses separate CLOB and auction networks with shared action encoders within each phase. A normalized ReLU state trunk produces an unrestricted scalar V_κ and query $u_\kappa \in \mathbb{R}^r$; a normalized encoder maps action coordinates to a bounded embedding $e_\kappa(a)$. The two architectures differ:

$$\begin{aligned} Q_{\theta_c}(\tilde{y}, a) &= V_c(\tilde{y}) + \frac{\langle u_c(\tilde{y}), e_c(a) \rangle}{\sqrt{r}} + \omega_c \mathbf{1}_{\{a \neq a_c^0\}}, \\ Q_{\theta_a}(\tilde{y}, a) &= V_a(\tilde{y}) + \frac{\langle u_a(\tilde{y}), e_a(a) - e_a(a_a^0) \rangle}{\sqrt{r}}. \end{aligned}$$

Here a_κ^0 is the phase no-op. The CLOB coefficient ω_c is learned from zero; the auction non-noop coefficient is fixed at zero, so its no-op value equals V_a . These coefficients are distinct from the preference weight ω_A . Other weights use ordinary initialization. The DQN action coordinates are $(v/V_{\text{max}}, \delta/L_{\text{max}})$ and $(K^a/(\beta K_{\text{max}}), \ell^a/B_{\text{max}}, c)$. Each state hidden layer has non-affine LayerNorm

before ReLU. The action encoder has two width- r linear layers, with ReLU after the first and non-affine LayerNorm followed by tanh after the second. Targets initially copy the online networks.

Training uses one-step masked Double DQN [52, 61]. Nonterminal targets use the next phase’s networks, including the auction continuation at the CLOB boundary; terminal targets contain no bootstrap. Raw observations determine masks and potentials, while replay stores transformed observations and conditioned rewards. The common structured-exploration period is followed by masked ε -greedy behavior; auction exploration mixes zero-slope controls with all admissible actions. Table 1 gives the hyperparameters. Algorithm 2 specifies updates and checkpoint selection for this architecture.

Algorithm 2 Phase-coupled masked Double DQN

Require: Hyperparameters in Table 1, frozen input transform, and selection protocol in Section 6.1

- 1: Initialize online/target phase networks, optimizers and replay buffers
 - 2: **for** each training episode permitted by the stopping rule **do**
 - 3: **for** each decision, with raw observation y in phase κ **do**
 - 4: Select a feasible action using structured exploration or masked ε -greedy behavior
 - 5: Observe y' , terminal indicator η and reward; compute $\tilde{R}$ using raw observations
 - 6: Store $(\mathcal{N}(y), a, \tilde{R}, \mathcal{N}(y'), \eta, \kappa', \text{Adm}(y'))$ in the current phase buffer
 - 7: **if** structured exploration is complete and the current buffer meets its threshold **then**
 - 8: Sample a minibatch; for each row set $Y = \tilde{R}$ if terminal, otherwise
 - 9: $a^* = \arg \max_{a' \in \text{Adm}(y')} Q_{\theta_{\kappa'}}(\mathcal{N}(y'), a')$, $Y = \tilde{R} + Q_{\theta_{\kappa'}}^-(\mathcal{N}(y'), a^*)$
 - 10: Take one AdamW step on the mean squared TD error, with cleared gradients and the tabulated norm bound
 - 11: Update only the current phase target: $\theta_{\kappa}^- \leftarrow (1 - \tau_{\text{targ}})\theta_{\kappa}^- + \tau_{\text{targ}}\theta_{\kappa}$
 - 12: **end if**
 - 13: **end for**
 - 14: Apply eligible joint validation and stopping
 - 15: **end for**
 - 16: **return** The joint checkpoint with the lowest eligible mean inventory-penalized shortfall
-

Terminal rewards already include settlement; terminal observations and masks are not used for continuation. At a CLOB–auction junction, κ' selects the auction networks. A run without an eligible checkpoint fails rather than returning an unvalidated policy.

4.3 Projected Continuous-Proposal Control.

DDPG, TD3 and SAC [42, 24, 32, 3] propose controls in $[-1, 1]^2$ during CLOB trading and $[-1, 1]^3$ during the auction. The environment executes $\Gamma_y(u)$, while replay retains the box-clipped proposal u , not the rounded executable order. The projection uses half-up rounding, inventory-capped CLOB sales, anchor-dependent auction price bounds and cancellation eligibility. It is nonsmooth and many-to-one; smooth-action convergence interpretations do not apply to the executable control map.

Write $\mathbf{r}(x) = \lfloor x + 1/2 \rfloor$. After clipping a proposal u to its box, CLOB execution uses

$$v = \min\{\lfloor I_t \rfloor, V_{\max}, \mathbf{r}((u_1 + 1)V_{\max}/2)\}, \quad \delta = \begin{cases} 0, & v = 0, \\ \mathbf{r}((u_2 + 1)L_{\max}/2), & v > 0. \end{cases}$$

Auction proposals give $K^a = \beta r((u_1 + 1)K_{\max}/2)$, $\ell_0 = r(B_{\max}u_2)$ and $c = \mathbf{1}_{\{u_3 \geq 0\}}C(X_t)$. For positive slope, clip ℓ_0 to the first and last integers in

$$\left\{ \ell \in \llbracket -B_{\max}, B_{\max} \rrbracket : |b^{\text{anc}}(y) + \ell| \leq B_{\infty}, \quad S_{\tau_{\text{op}}}^{\text{mid}} + \alpha(b^{\text{anc}}(y) + \ell) \geq 0 \right\}.$$

If this set is empty, set slope and offset to zero; zero slope always uses zero offset. Cancellation eligibility is unchanged by this fallback. The anchor is the indicative or frozen-mid anchor defined in Section 3. Replay retains the box-clipped proposal before these operations. No auction inventory cap is imposed. We use Stable-Baselines3 2.7.1 with separate phase models and ordinary initialization. DDPG adds affine LayerNorm before each critic hidden-layer activation; its actor is unchanged. TD3 and SAC use twin critics without this additional normalization. Auction opening is not economic termination.

For continuous learners, let μ_a^- denote the auction target actor and Q_a^- (or $Q_{a,j}^-$) its target critic(s). The auction continuation at a sampled CLOB junction is

$$\begin{cases} Q_a^-(\tilde{y}', \mu_a^-(\tilde{y}')), & \text{DDPG,} \\ \min_j Q_{a,j}^-(\tilde{y}', u'), & \text{TD3,} \\ \min_j Q_{a,j}^-(\tilde{y}', u') - \zeta_a \log \pi_a(u' | \tilde{y}'), & \text{SAC.} \end{cases}$$

TD3 adds clipped Gaussian target noise to its target proposal and clips the result to the proposal box. SAC samples its current actor and uses its current positive entropy coefficient ζ_a . The continuation is added at sampling time, with further CLOB bootstrapping suppressed. Actual terminal rows have no continuation. DDPG/TD3 actor updates maximize the first critic; SAC includes its entropy term. The critic-loss aggregation and actor/target update periods are given in Table 1. DDPG’s actor rate is applied by its optimizer hook; SAC’s temperature rate follows the same fixed episode decay as its network rates.

Optimizer hooks impose the actor and critic gradient bounds in Table 1; the SAC temperature optimizer is not clipped by these hooks. SAC adjusts entropy automatically in each phase. After common structured exploration, DDPG/TD3 use Gaussian proposal noise and SAC samples its stochastic policy. Evaluation uses deterministic actors, including the squashed Gaussian mean for SAC. All learners return the joint checkpoint selected under Section 6.1.

5 Stylized Reference Benchmarks

We compare the performance of our DQN-learned policy and the continuous-control extensions against two different stylized reference benchmarks for optimal market making. Both benchmarks serve an illustrative purpose of closed-form approaches as we primarily compare the DQN policy against the continuous-control extensions. The first benchmark is adapted from the optimal market making models of Avellaneda and Stoikov [4] later solved explicitly by Guéant, Lehalle and Fernandez-Tapia [31]. We adapt the market making model to liquidation only and without risk aversion for the continuous phase. We then suggest an approximation allowing a straightforward application to our discrete-time setup. The second benchmark is the time weighted average price policy for the continuous phase. We adopt the same heuristic liquidation rule for the auction phase. These benchmarks are external reference policies for optimal liquidation and are not required to belong to the learned policy class II. A theoretical solution to the optimal market making problem on a closing auction is left for future work.

Setting	DQN	DDPG	TD3	SAC
Backend	Native		Stable-Baselines3 2.7.1	
Replay capacity (per phase)			50,000	
Replay minima N_w^c/N_w^a			512/512	
Structured exploration	Episodes 0–31; no optimizer updates			
Minibatch size	128			
Hidden widths (ReLU)	256, 256			
Replay reward scale c_r	1			
One-step continuation factor	1			
Critic updates per transition	At most one, after both learning gates			
Optimizer	AdamW	Adam	Adam	Adam
Weight decay	10^{-4}	0	0	0
Gradient-norm bound	10	1	1	1
Action-embedding rank	32	–	–	–
<i>Initial rates; multiplied by $\max\{0.1, 2^{-e/90}\}$ in episode e</i>				
Critic	1.5×10^{-4}	3×10^{-4}	3×10^{-4}	3×10^{-4}
Actor	–	10^{-4}	3×10^{-4}	3×10^{-4}
Temperature	–	–	–	3×10^{-4}
Number of online critics	1	1	2	2
Critic loss	MSE	MSE	Sum of MSEs	Half-sum of MSEs
Polyak coefficient	0.005	0.0025	0.005	0.005
Target-update period [†]	1	1	2	1
Actor-update period [†]	–	1	2	1
<i>Behavior after structured exploration; buffer fallback specified in text</i>				
Behavior	Masked mixture	Gaussian	Gaussian	Stochastic
DQN epsilon endpoints	$1 \rightarrow 0.05$	–	–	–
DQN decay start/end (e)	5/95	–	–	–
Auction zero-slope mixture	0.5	–	–	–
Proposal-noise std.	–	0.1	0.1	–
TD3 target-noise std./clip	–	–	0.2/0.5	–
Initial entropy coefficient	–	–	–	0.001 (auto)
Target entropy (CLOB/auction)	–	–	–	–2/–3

Table 1: Learner hyperparameters. Target and actor periods ([†]) count updates of the relevant phase critic, not market decisions. Gradient bounds for the continuous learners apply separately to actor and critic, not to SAC temperature. Input normalization and architectural LayerNorm are specified in the text. Rates decay with the global episode index, without restarting after exploration. Selection uses inventory-penalized shortfall (Section 6.1).

5.1 Avellaneda-Stoikov Optimal Market Making

In this section, we will recall the main results of [4] and [31], in the case where the market maker acts as a seller only. We consider the continuous phase, that is $t < \tau^{\text{op}}$ in what follows. For this auxiliary continuous-time benchmark, we work on $[0, T]$ with the chosen horizon $T = t_n$, where $t_n = \tau^{\text{op}} - 1$ is the final CLOB decision time in our initial framework. This benchmark horizon is

distinct from the end of continuous execution at τ^{op} . The market mid price is assumed to follow an arithmetic Brownian motion $dS_t^{\text{mid}} = \sigma dB_t$ with $\sigma > 0$. We assume that transactions have constant size Δ . For simplicity, assume $\Delta = 1$. Fix an integer inventory cap $Q \in \mathbb{N}$ and initial inventory $q_0 = I_0 \in \{0, \dots, Q\}$. The inventory process is $q_t = q_0 - N_t^a$, where N^a is the controlled counting process giving the cumulative number of shares sold by the market maker. Formulated initially by Avellaneda and Stoikov, we assume that the intensity of N^a depends on the spread $\delta_t^a = k_t^\bullet - k_t^{\text{mid}}$ as $\lambda_t^a(\delta_t^a) = \mathbf{1}_{\{q_{t-} > 0\}} A e^{-\alpha k \delta_t^a}$ for $A, k > 0$. To specify the driving noise, let $\mathcal{M}^a$ be a Poisson random measure on $[0, T] \times \mathbb{R}_+$ with compensator $ds dz$, independent of B . Let $\mathbb{F}^{\text{AS}}$ be the usual augmentation of the filtration jointly generated by B and $\mathcal{M}^a$ up to time t . Prices and execution counts are adapted to this common filtration, and quote controls are predictable with respect to it. Executions are generated by

$$\forall t \in [0, T], \quad N_t^a = \int_{(0, t] \times \mathbb{R}_+} \mathbf{1}_{\{z \leq \lambda_s^a(\delta_s^a)\}} \mathcal{M}^a(ds, dz).$$

The primitive noises are independent, but the controlled count N^a need not be independent of B when quotes respond to the price history. The cash process X_t of the market maker evolves according to $dX_t = (S_t^{\text{mid}} + \alpha \delta_t^a) dN_t^a$. Let $T = t_n$ and let $\tilde{\mathcal{A}}$ consist of bounded $\mathbb{F}^{\text{AS}}$ -predictable quote controls while $q_{t-} > 0$, with no order posted and no further executions at $q = 0$. The stopped intensity keeps inventory in $\{0, \dots, Q\}$. The market maker optimizes

$$(\mathbf{M}) \quad \sup_{\delta^a \in \tilde{\mathcal{A}}} \mathbb{E} \left[X_T + q_T S_T^{\text{mid}} \right]$$

Proposition 5.1. *For $t \in [0, T]$ and $q \in \{1, \dots, Q\}$, the optimal quotes solving $(\mathbf{M})$ are given by*

$$\delta^{a,*}(t, q) = \frac{1}{\alpha k} \left[1 + \ln \left(\frac{v_q(t)}{v_{q-1}(t)} \right) \right],$$

where

$$\forall q \in \llbracket 0, Q \rrbracket, \quad v_q(t) = \sum_{j=0}^q \frac{(Ae^{-1}(T-t))^j}{j!}.$$

The boundary conditions are $v_0(t) = 1$ for $0 \leq t \leq T$ and $v_q(T) = 1$ for $0 \leq q \leq Q$. At $q = 0$ the policy abstains; the quote formula is not evaluated. At time $t = T$, the final quote uses the continuous extension $\delta^{a,*}(T, q) = 1/(\alpha k)$ for $q \geq 1$, which follows from $v_q(T) = v_{q-1}(T) = 1$.

Proof. Note that $(\mathbf{M})$ corresponds to the risk-neutral market maker $\gamma \rightarrow 0$ in the standard optimal market making problem investigated in [4, 31]. In this case, one proves that the problem reduces to solving the ODE system

$$\forall q \in \llbracket 1, Q \rrbracket, \quad v_q'(t) = -Ae^{-1} v_{q-1}(t),$$

and we directly get optimal quotes from [31] when $\gamma \rightarrow 0$ by

$$\delta^{a,*}(t, q) = \frac{1}{\alpha k} \left[1 + \ln \left(\frac{v_q(t)}{v_{q-1}(t)} \right) \right].$$

We prove the result by induction. Note that $v_0 = 1$. Assume now that

$$\forall q \geq 1, \quad v_{q-1}(t) = \sum_{j=0}^{q-1} \frac{(Ae^{-1}(T-t))^j}{j!}.$$

Since $v_q(T) = 1$, we compute

$$\begin{aligned} v_q(t) &= v_q(T) - \int_t^T v'_q(s) ds = 1 + Ae^{-1} \int_t^T v_{q-1}(s) ds = 1 + Ae^{-1} \sum_{j=0}^{q-1} \int_t^T \frac{(Ae^{-1}(T-s))^j}{j!} ds \\ &= 1 + \sum_{j=0}^{q-1} \frac{(Ae^{-1})^{j+1}}{j!} \int_0^{T-t} s^j ds = 1 + \sum_{j=0}^{q-1} \frac{(Ae^{-1}(T-t))^{j+1}}{(j+1)!} = \sum_{j=0}^q \frac{(Ae^{-1}(T-t))^j}{j!}. \end{aligned}$$

This completes the proof by induction. The rest follows. $\square$

The quote formula applies only for $q \in \{1, \dots, Q\}$; the zero-inventory state has the no-trading boundary specified above. The continuous-time derivation assumes unit-sized transactions and controls the quote offset $\delta^{a,*}(t, q)$, measured in ticks, rather than a separately submitted volume. Before grid projection, the corresponding quote price is $S_t^{\text{mid}} + \alpha \delta^{a,*}(t, q)$. At each CLOB decision time $t = t_i$, the agent submits $v_t = V_{\max} \wedge \lfloor q_t \rfloor$ units. If $v_t > 0$, the quote offset δ_t is obtained by evaluating $\delta^{a,*}(t, 1 \vee \lceil q_t \rceil)$ and projecting it to the nearest admissible integer offset. The submitted price $S_t^\bullet = S_t^{\text{mid}} + \alpha \delta_t$ remains fixed throughout $(t_i, t_{i+1}]$: all fills of that order occur at this price, rather than at quotes recomputed as time and inventory change. If $v_t = 0$, the agent submits no order, uses the canonical offset $\delta_t = 0$, and does not evaluate the quote formula.

This approximation therefore fixes the quote between decisions and imposes an explicit submitted-volume cap. The agent does not necessarily expose her whole inventory: the cap can limit execution even when sufficient buying volume reaches her price. Unlike the continuous-time unit-fill model, the simulator permits partial and multi-unit fills, whose total cannot exceed the submitted volume; any unexecuted remainder expires at the next decision time.

Once the auction opens, an inventory $q_{\tau^{\text{op}}} \geq 0$ remains. We implement the following heuristic policy. Let $\mathcal{E} = \{i \in \{0, \dots, n\} : E_{t_i} > 0\}$ be the CLOB decisions with positive execution. Each such decision contributes its submitted price once, regardless of the number of shares executed. For $\mathcal{E} \neq \emptyset$, define

$$\tilde{S} = \frac{1}{2} \left(\frac{1}{|\mathcal{E}|} \sum_{i \in \mathcal{E}} S_{t_i}^\bullet + \max_{i \in \mathcal{E}} S_{t_i}^\bullet \right).$$

If no CLOB execution occurred, we set $\tilde{S} := H_{t_n}^{\text{cl}}$. The whole remaining inventory $q_{\tau^{\text{op}}}$ is put on the auction at $\tilde{S}$, with supply function $\tilde{g}_z(p) = \min(q_{\tau^{\text{op}}}, \min(zq_{\tau^{\text{op}}}, \beta K_{\max})(p - \tilde{S})_+)$. In Section 6, we consider $z = 10$. This single order is submitted right at the beginning of the auction, and only potentially executed at the clearing time. If $q_t = 0$, the benchmark submits no CLOB order; the quote formula is used only for $q_t \geq 1$. For both AS and TWAP, the strategic contribution to aggregate net supply is replaced by the capped, one-sided schedule $\tilde{g}_z$. The continuous clearing root is computed using this schedule, which is also evaluated at the bracketing ticks when determining matched volume. The same tick-selection and pro-rata allocation rules then apply. In particular, the benchmark's executed sale cannot exceed its auction-opening inventory.

5.2 Time-Weighted Average Price Benchmark

The second benchmark is deliberately simpler. Given the current inventory q_t , the trader submits a deterministic volume $v_t = V_{\max} \wedge \lfloor q_t \rfloor \wedge \lceil q_t / (T - t + 1) \rceil$, quoted at the best ask price, *i.e.* $\delta_t = 1$. Such a policy does not guarantee full liquidation during the continuous trading phase, as execution is conditional on order matching. For the auction phase, we adopt exactly the same liquidation policy as in the Avellaneda–Stoikov benchmark.

6 Numerical Simulations

This section employs the generative stochastic market model formulated in Section 2 to simulate continuous trading and closing auctions. We compare the DQN-learned policy against the learned continuous-control methods and the two stylized reference benchmarks: Avellaneda-Stoikov (AS) strategy, see Section 5.1 and the TWAP strategy see Section 5.2. The goal of this section is to compare the DQN policy with the projected continuous-control policies and the stylized reference benchmarks under the same simulator configuration. We start by generating an emulator of the CLOB followed by the closing auction. The algorithm used to generate the market mechanism is defined in Section 6.2 with Algorithm 3. In Section 6.3 we generate mid price process values by using the rough-Heston to describe the evolution of the price of a risky asset, see [28].

In both simulations we consider the following setting: a two-hour CLOB phase followed by a 30-minute closing auction, with the mid price generated by the rough Heston model or with historical mid-price input. This corresponds roughly to what happens at exchanges like Xetra and the CLOB phase can be viewed as the trading sessions leading up to the closing auction. The 30-minute auction duration here is however a stylized numerical choice, not a calibrated exchange duration.

6.1 Experimental Protocol

We evaluate DQN, DDPG, TD3 and SAC in six price environments: synthetic rough-Heston paths and historical midprice paths for CAT, GOOGL, JPM, MSFT and PG. Each learner is trained with ten master seeds under each specification. The main experiment comprises 240 runs under the headline specification and 200 runs for synthetic controls. Two subsequent synthetic comparisons add 40 runs each, using the rerun headline checkpoints as their common reference. AS and TWAP require evaluation but no training.

All numerical exhibits below use the updated 520-run v20 bundle with volume-maximizing tick projection for both lagged auction indications and terminal settlement. Its 320 synthetic runs were replaced using the internal rough-Heston grid with maximum step 0.25 minutes described below. Synthetic forecast weights, normalization and reference conditioning were refitted on training data, policies were retrained and reselected, and matched evaluations were repeated. The 200 historical runs and their forecast fit retain their original artifacts and source revisions. Neither the superseded decision-grid synthetic results nor the earlier nearest-tick campaign are pooled into these exhibits. The replication bundle binds the replacement and retained historical artifacts to saved configurations, hashes and source disclosures. Differences from the superseded synthetic campaign include recalibration and retraining; they are not fixed-policy, coupled-mesh estimates of discretization error, and the selected step is not an established accuracy threshold.

Historical training draws whole sessions from all five tickers over August 3–14, 2026. Validation and testing are conducted separately for each ticker over August 17–21 and August 24–28, respectively. No test episode enters the stated forecast fitting, normalization, or within-run validation-checkpoint rule. The test dates were inspected during development, so the historical results should not be interpreted as evidence from an untouched holdout. Repeating these dates with new simulation seeds measures variation in the simulated market, not variation across additional historical sessions.

We construct the historical paths from Alpaca SIP bid–ask quotes recorded between 13:30 and 16:00, New York time. Quotes are excluded if their timestamps are invalid, their prices or sizes are nonfinite or nonpositive, or the bid exceeds the ask; locked quotes are retained. At each minute, we take the most recent valid quote at or before that timestamp and compute the midpoint of its bid and ask. Dataset construction fails if a required quote is missing or more than 60 seconds old.

Each session is rebased to an initial midprice of 100. All historical midprices are projected onto the tick grid after rebasing, using nearest-tick rounding with half-ticks rounded upward, as before. Between minute observations, the simulator uses the last observed midpoint: at time t , measured in minutes from 13:30, it takes observation $\lfloor t \rfloor$ without interpolation. The 15:30 midpoint remains fixed throughout the simulated auction, and later historical quotes are not used.

Forecasts are fitted using training data only, as described in Section 2.1.4; input transformations and reference values follow Section 4.1. Calibration, training, validation and testing use separate simulation streams. Within each reported run, test outcomes enter neither these calibration fits nor validation-checkpoint selection; the historical test dates were nevertheless inspected during development.

Training is limited to 800 episodes per run. Once the learning conditions in Table 1 are met, we evaluate policies on a fixed set of 128 validation episodes every 50 completed training episodes. We also evaluate at the end of the first episode in which each active phase has accumulated 2,000 optimizer updates. This update threshold determines eligibility for policy selection; for runs without auction access, it applies only to the CLOB phase.

Among eligible evaluations, we retain the joint CLOB–auction checkpoint with the lowest mean inventory-penalized shortfall, IS_λ . A checkpoint is replaced only by a strict improvement, so ties favor the earlier policy. Training stops after four consecutive eligible evaluations without improvement, or at the episode limit. The final model is eligible only if it has undergone eligible validation. Untrained policies are evaluated for diagnostic purposes and do not enter checkpoint selection.

Each selected policy is evaluated deterministically on 100 test episodes. Learned policies, untrained models, and the AS and TWAP references face the same episode and environment-seed pairs. We preserve this pairing across learners and training specifications within each master seed and market. In the historical setting, the 100 episodes reuse the five test dates with different simulated order-flow and auction draws.

Table 2 defines the eight synthetic training schemes. Historical experiments use only the headline specification, J-I. The common learning adjustments are initial-value centering, CLOB potential credit and the reference adjustment; dense and sparse auction credit are defined in Section 4.1. All schemes with auction access allow signed schedules and retain cancellation eligibility. Without auction access (No-A), residual inventory is marked and penalized at auction opening rather than forcibly executed. Shortfall is then computed with zero auction fill and the opening-time residual position.

The main synthetic controls distinguish five effects: dense rather than sparse auction credit (E-F versus E-S), access to the H forecast feature (E-F versus E-0), indicative rather than frozen-mid anchoring (J-I versus J-F), combined preferences at a fixed anchor (J-F versus E-F), and auction access after retraining (E-0 versus No-A). Removing H does not remove the other observations that may contain information about clearing. We also decompose the contribution of realized auction trades into execution value and inventory-risk relief, relative to submitting no auction orders on the same CLOB trajectory. Unlike the retrained comparison, this accounting exercise holds continuous trading fixed; it does not equalize auction-opening inventories across learners.

The two subsequent comparisons separate the full training scheme from the role of the added preferences. J-I versus Cash-I compares the headline scheme with training based on actual cash flows, fees, residual marking and terminal inventory risk, without preferences, initial-value centering, or potential and reference adjustments. J-I versus E-I removes only the preferences, holding H access, anchoring and conditioning fixed. This second comparison identifies the effect of the combined

preference terms, not their individual contributions. We do not conduct corresponding historical ablations, separate cancellation controls, or experiments in which all learners share the AS auction policy.

Scheme	Campaign	H	Anchor	Preferences	Conditioning	Auction
J-I	Main	On	Indicative	On	Common + dense	On
J-F	Main	On	Frozen mid	On	Common + dense	On
E-F	Main	On	Frozen mid	Off	Common + dense	On
E-S	Main	On	Frozen mid	Off	Common + sparse	On
E-0	Main	Off	Frozen mid	Off	Common + dense	On
No-A	Main	Off	–	Off	Common, CLOB only	Off
Cash-I	Follow-up 1	On	Indicative	Off	None	On
E-I	Follow-up 2	On	Indicative	Off	Common + dense	On

Table 2: Synthetic training schemes. Each row contains four learners and ten seeds. H denotes access to the forecast feature; common conditioning includes centering, CLOB potential credit and the reference adjustment. All schemes retain network-input transformations.

We express shortfall in basis points of initial notional by multiplying model cash amounts by $10^4/(S_0^{\text{mid}}I_0)$. For each policy, we average the 100 test episodes within each master seed and then give the ten seed means equal weight. Policy comparisons are paired at the episode level before averaging within seeds. A positive reduction denotes lower shortfall for the policy of interest. We compare training schemes using shortfall rather than accumulated training rewards, whose centering differs across specifications (Section 4.1).

Uncertainty is measured across training seeds, not by treating test episodes as independent training replications. We construct pointwise 95% percentile intervals from 10,000 bootstrap resamples of the ten seed means, or of the ten paired seed differences for comparisons, using the 2.5th and 97.5th percentiles. All seeds are retained. Historical summaries first average the five tickers equally within each seed and then resample whole seed blocks, preserving the dependence across tickers. Historical inference is therefore conditional on these tickers and the reused dates. Learning-curve bands are calculated only at training budgets observed for every seed, without extrapolation beyond early stopping. Unless explicitly omitted, intervals in the results and figures follow these conventions; they are pointwise and are not adjusted for multiple comparisons.

6.2 Generative Stochastic Market Model

We now explain in detail our market emulator to generate the financial market in our setting. The algorithm is described in Algorithm 3. It specifies only the exogenous random variables and accepted exogenous book updates. More precisely, we say that a candidate exogenous linear auction book is valid if the aggregate slope $G_t \geq D_\mu$ for a numerical lower slope bound $D_\mu > 0$ and there is a nonnegative continuous clearing root, as in Proposition 2.1.

At each auction decision time, the accepted exogenous update returned by Algorithm 3 is incorporated into the current configuration before the agent acts via the following chronology.

1. If $k = n + 1$, set $H_{t_{n+1}}^{\text{cl}} = H_{t_n}^{\text{cl}}$. If $k \geq n + 2$, compute $H_{t_k}^{\text{cl}}$ from the complete post-action book stored at t_{k-1} .

Algorithm 3 Exogenous stochastic market generator

Require: Parameter vector $(\lambda_0, v_m, \gamma_m, V, \beta_a, \beta_b, V_\infty, \rho_{\text{lob}}, U_1, U_2, L_\infty, D_\mu, B_\infty, p_1, p_2, p_3, p_4)$

```
1: Sample the independent CLOB counting processes  $N^+, N^- \sim \text{PP}(\lambda_0)$  on  $[0, \tau^{\text{op}}]$ 
2: Construct the complete realized grid  $(t_i)_{0 \leq i \leq m+1}$  according to Definition 3.1
3: for  $k = 0, \dots, n$  do
4:   for  $\zeta \in \{+, -\}$  do
5:     Sample  $V_{t_k}^{\zeta, 1} \sim V_\infty \text{Beta}(\beta_a, \beta_b)$ 
6:     for  $\ell = 1, \dots, L_\infty - 1$  do
7:       Set  $V_{t_k}^{\zeta, \ell+1} = \rho_{\text{lob}} V_{t_k}^{\zeta, \ell}$ 
8:     end for
9:     for  $r = N_{t_k}^\zeta + 1, \dots, N_{t_{k+1}}^\zeta$  do
10:      Sample  $Z_{k,r}^\zeta \sim \text{Pareto}(v_m, \gamma_m)$  and set  $\nu_{k,r}^\zeta = \min\{Z_{k,r}^\zeta, V\}$ 
11:    end for
12:  end for
13:  Send exogenous CLOB snapshot and arrivals on  $(t_k, t_{k+1}]$  to the controlled environment
14: end for
15: Receive  $O_{\tau^{\text{op}}}^*$  from the controlled environment, recompute the residual-book replacement moments without updating the observed forecast, initialize the carry-over schedules  $\mathcal{B}_{\tau^{\text{op}}}^* = \{g_{k, \tau^{\text{op}}}^* : k \in \mathcal{K}_*\}$  and set  $H_{t_{n+1}}^{\text{cl}} \leftarrow H_{t_n}^{\text{cl}}$ 
16: for  $k = n + 1, \dots, m$  do
17:   Sample proposed auction updates  $B_{t_k} \sim \mathcal{B}(p_1)$ ,  $D_{t_k} \sim \mathcal{B}(p_2)$ ,  $J_{t_k}^\pm \sim \mathcal{B}(p_3)$ ,  $G_{t_k}^\pm \sim \mathcal{B}(p_4)$ 
18:   if  $B_{t_k} = 1$  then
19:     Sample  $K_{t_k}^r \sim \mathcal{U}([U_1, U_2])$ ,  $S_{t_k}^r \sim S_{\tau^{\text{op}}}^{\text{mid}} + \alpha \mathcal{U}([-B_\infty, B_\infty])$  and add the new exogenous schedule only if the resulting exogenous book remains valid
20:   end if
21:   if  $D_{t_k} = 1$  and a nonpersistent exogenous schedule is present then
22:     Propose canceling one uniformly selected nonpersistent live exogenous schedule and accept the cancellation only if the resulting exogenous book remains valid
23:   end if
24:   for  $\zeta \in \{+, -\}$  do
25:     if  $J_{t_k}^\zeta = 1$  then
26:       Sample  $Z_{t_k, \zeta} \sim \text{Pareto}(v_m, \gamma_m)$  and set  $\nu_{t_k, \zeta} = \min\{Z_{t_k, \zeta}, V\}$ 
27:       Add the corresponding market order if the resulting book is valid
28:     end if
29:     if  $G_{t_k}^\zeta = 1$  and an active market order is present on side  $\zeta$  then
30:       Cancel a uniformly sampled order if the resulting book is valid
31:     end if
32:   end for
33: end for
```

2. Sample the exogenous proposals at t_k and accept each proposal only if the resulting exogenous book remains valid.
3. Construct the pre-action configuration X_{t_k} and reveal $\varphi(X_{t_k})$ to the agent.
4. If $c_{t_k} = 1$, remove every still-active agent schedule submitted strictly before t_k .
5. Add the current schedule $p \mapsto K_{t_k}^a(p - S_{\tau^{\text{op}}}^{\text{mid}} - \alpha b_{t_k}^a)$.

6. Store the resulting post-action book. If $k = m$, freeze this book, process no event on $(t_m, \tau^{\text{cl}}]$, and apply the terminal clearing and allocation rule.

After the final action at t_m , the controlled environment freezes the post-action book and applies the clearing and allocation rule of Section 2.1.3.

The emulator parameters are

$$(\lambda_0, v_m, \gamma_m, V, \beta_a, \beta_b, V_\infty, \rho_{\text{lob}}, U_1, U_2, B_\infty, D_\mu, L_\infty, p_1, p_2, p_3, p_4).$$

- λ_0 represents the intensity of the counting processes N^+ and N^- on the continuous phase, modeled as independent Poisson processes, that are the arrivals per side per minute. They are sampled in the first step of the algorithm.
- (v_m, γ_m) are the parameters of the Pareto distribution modeling the volumes of the market orders throughout the whole trading sessions. This reproduces the well-known heavy tail behavior of the density of market order size, see for example [29, 25, 10]. The market model allows orders of maximum size V : greater liquidity takers will accumulate at volume V .
- (β_a, β_b) represent the parameters of the Beta distribution modeling the first volume of the limit order book. The exogenous best-level depth law $V_\infty \text{Beta}(\beta_a, \beta_b)$ is a stylized modeling choice, not an empirically fitted distribution. The beta policies in [35, 63] instead parameterize a market maker's allocation of its own order volume across price levels; they provide an analogy, not an estimation or validation of this exogenous depth law. Note that V_∞ is a rescaling parameter as the Beta distribution has support $[0, 1]$. Further volumes in the limit order book decay geometrically with parameter $\rho_{\text{lob}} \in (0, 1]$ and are truncated via the maximum order book depth L_∞ , that is $V_t^{\zeta, j} = 0$ for $j > L_\infty$.
- (U_1, U_2) represent the bounds between which exogenous market maker sample their supply slopes from during the auction phase.
- B_∞ represents the integer bound of the price at which exogenous market makers quote during the auction phase. Exogenous market makers during the auction have reference prices sampled as $S_t^i \sim S_{\tau_{\text{op}}}^{\text{mid}} + \alpha \mathcal{U}(\llbracket -B_\infty, B_\infty \rrbracket)$ because the auction opening time is the last time market participants see the mid price. Note that a similar auction reference-price specification has been introduced in [20].
- At each one-minute auction decision, independent Bernoulli proposals govern market-maker arrival (p_1), cancellation of a nonpersistent market-maker schedule (p_2), and, independently on each side, market-order arrival (p_3) and cancellation of an active market order (p_4). Each proposed update is accepted only if the resulting exogenous book is valid.

The proposal variables and their marks are sampled independently. The accepted book updates are state-dependent because of the validity screen. We do assume that the limit order book is refreshed from one time step to the next one.

Initial price and inventory are both rebased to 100, giving initial notional 10,000 model cash units. A tick of 0.01 need not equal one cent in an unrebased historical price. Cash and shortfall have units of model price times model inventory; reported shortfall is in basis points of initial notional. The physical slope bound $\beta K_{\text{max}} = 8$ applies to one schedule, not to aggregate live slope, signed fills or terminal inventory. Exogenous schedules converted from the residual CLOB retain their original reference levels and are not clipped to B_∞ . The local strategic offset bound B_{max} is distinct from

Symbol	Value	Meaning in model units
$[t]$	minutes	Unit of the simulator clock
τ^{op}	120	Auction opening time
τ^{cl}	150	Clearing time; 30 auction decisions
S_0^{mid}	100	Rebased initial midprice
I_0	100	Initial model inventory
λ_0	0.5/min/side	CLOB Poisson intensity per side
v_m	2	Pareto scale (model inventory units)
γ_m	2.5	Pareto shape
V_∞	2	Top-depth Beta scaling
β_a, β_b	2, 5	Top-depth Beta shapes
ρ_{lob}	0.96	Geometric depth decay
V	30	Exogenous market/taker-order volume cap
V_{max}	30	Strategic CLOB submitted-volume cap
L_∞	200	Exogenous LOB depth in levels
L_{max}	12	Strategic CLOB quote-offset bound in ticks
B_∞	150	Reference-price deviation bound for strategic schedules and new exogenous arrivals, in ticks from frozen mid
B_{max}	10	Local strategic offset half-width around the selected anchor, in ticks
D_μ	0.1	Lower bound on valid exogenous aggregate slope
U_1, U_2	1, 20	Exogenous per-schedule slope support endpoints
p_1	1	Market-maker arrival proposal probability per auction minute
p_2	0	Exogenous market-maker cancellation proposal probability
p_3	0.3	Market-taker arrival proposal probability per side per minute
p_4	0.05	Active market-order cancellation proposal probability per side per minute
λ	0.01	Terminal quadratic inventory-risk coefficient
q	0	Purchase attenuation only; other preferences remain active
k^*	10,000	Active CLOB preference clipping scale
ω_A	10^{-4}	Active fictive auction-credit weight
d	0.001	Stylized withdrawal-cost increment, $d_{t_j} = (j - n - 1)d$
α	0.01	Tick in rebased model-price units
β	0.25	Strategic slope quantum (inventory units per price unit)
K_{max}	32	Maximum integer slope index, with zero included
βK_{max}	8	Maximum physical slope of one strategic schedule in model units
H_0	100	Initial raw-recursion clearing signal
η_H	0.95	Raw clearing-price smoothing weight in Algorithm 1

Table 3: Shared simulator, action-grid and preference parameters for the synthetic and historical headlines.

this absolute reference-price bound. Order-flow intensities, depth distributions, auction slopes and withdrawal costs are stylized calibration choices, not estimates of exchange liquidity or an exchange

fee schedule. Historical inputs supply only rebased midprice paths within this simulated market.

6.3 Rough Heston Model for the Mid Price: Generative Data Approach

In the first numerical implementation, we assume that the mid price S_t follows a rough Heston model. The motivation is based on the so-called rough volatility of financial assets [28, 12, 2]. Consider $\rho \in [-1, 1]$ a constant, $S_0 = 100$ (which is the numerical value we work with in this section), V_0 , θ , ς and ν positive constants, and $0 < H < 1/2$ for the rough-volatility regime. Recalling [28, 57], the rough Heston model writes

$$\begin{cases} S_t = S_0 + \int_0^t S_s \sqrt{V_s} d(\rho B_s + \sqrt{1 - \rho^2} B_s^\perp), \\ V_t = V_0 + \int_0^t K_H(t-s) ((\theta - \varsigma V_s) ds + \nu \sqrt{V_s} dB_s), \end{cases}$$

where $(B, B^\perp)$ are two independent Brownian motions and $K_H(t) = t^{H-1/2}/\Gamma(H+1/2)$ for $t > 0$.

Set $Y_0 = \log S_0$. We use the Euler-type scheme of [57] on a refined price-simulation grid. For each episode, we first construct the realized CLOB decision grid $0 = t_0 < \dots < t_n = \tau^{\text{op}} - 1 < t_{n+1} = \tau^{\text{op}}$. We subdivide each interval $[t_i, t_{i+1}]$ into the smallest power-of-two number of equal subintervals whose lengths do not exceed $\delta_{\text{int}} = 0.25$ minutes (15 seconds). Their endpoints form the refined grid $0 = u_0 < \dots < u_N = \tau^{\text{op}}$ with $\max_{0 \leq i < N} (u_{i+1} - u_i) \leq \delta_{\text{int}}$ which contains every CLOB decision time and auction opening. This construction gives nested grids as the step bound is tightened.

Remark 6.1. *We use a maximum internal step of $\delta_{\text{int}} = 0.25$ minutes to increase temporal resolution while keeping repeated policy training computationally feasible. While coupled refinement checks indicate residual mesh sensitivity, finite-grid accuracy has not, however, been established. Accordingly, the synthetic results are conditional on this discretized simulator. Robustness of the reported policy comparisons to further mesh refinement has not been established.*

We measure model time in trading years: for $s^* = 252 \cdot 6.5 \cdot 60$, set $\bar{u}_i = u_i/s^*$ and $\Delta \bar{u}_{i+1} = \bar{u}_{i+1} - \bar{u}_i$. Using Brownian drivers independent of the order-flow processes, we simulate the rough-Heston approximation conditional on the complete refined grid. Write $\Delta B_{i+1} = B_{\bar{u}_{i+1}} - B_{\bar{u}_i}$ and $\Delta B_{i+1}^\perp = B_{\bar{u}_{i+1}}^\perp - B_{\bar{u}_i}^\perp$ for $i \in \llbracket 0, N-1 \rrbracket$. Conditional on the grid, these increments are independent across intervals and drivers, centered Gaussian, and have variance $\Delta \bar{u}_{i+1}$. For $k \in \llbracket 1, N \rrbracket$, the log-price and variance approximations at u_k are

$$\begin{cases} Y_k = Y_0 + \sum_{i=0}^{k-1} \left[-\frac{1}{2} (V_i)_+ \Delta \bar{u}_{i+1} + \sqrt{(V_i)_+} \left(\rho \Delta B_{i+1} + \sqrt{1 - \rho^2} \Delta B_{i+1}^\perp \right) \right], \\ V_k = V_0 + \sum_{i=0}^{k-1} K_H(\bar{u}_k - \bar{u}_i) \left[(\theta - \varsigma (V_i)_+) \Delta \bar{u}_{i+1} + \nu \sqrt{(V_i)_+} \Delta B_{i+1} \right]. \end{cases}$$

The variance sum retains all preceding refined-grid points; it is not restarted at a trading decision. Computed negative variance values are retained, with their positive parts used only in the coefficients. We use the illustrative parameters $H = 0.1$, $\rho = -0.7$, $V_0 = 0.02$, $\theta = 0.02$, $\varsigma = 0.3$ and $\nu = 0.3$, adapted from [1].

Let $\kappa(i)$ satisfy $u_{\kappa(i)} = t_i$. The trading environment observes the price only at the original decision times and auction opening, with $S_{t_i}^{\text{mid}} = \alpha \lfloor \exp(Y_{\kappa(i)})/\alpha + 1/2 \rfloor$. The midprice is simulated through $u_N = \tau^{\text{op}}$ and then frozen at $S_{\tau^{\text{op}}}^{\text{mid}}$ throughout the auction. The opening forecast inherits the last pre-action CLOB forecast.

Now, one may notice that the AS stylized reference benchmark assumes that $dS_t = \sigma dB_t$, instead of the rough Heston model. In this sense, the goal of the numerical simulation is to compare how our deep reinforcement learning policies perform against a stylized reference benchmark, without claiming optimality of this benchmark. The benchmark is deliberately stylized and is not optimal for the simulated rough-Heston environment; it is therefore interpreted as a reference policy rather than as an oracle comparator.

As σ drops out of the risk-neutral $\gamma \rightarrow 0$ policy used here, we omit its calibration. Following the proportional power-law argument in [4], we adopt the heuristic parameterization $A = \lambda_0/\gamma_m$ and $k = \gamma_m \hat{K}$. Note that A is per minute. Here, $\hat{K}$ is the proportionality constant in the empirical relationship $\hat{K}\Delta p = \ln(Q)$, where Δp is the move in price when a market order of size Q arrives. We did a least squares regression to determine $\hat{K}$, by attempting 5,000 simulated samples of the limit order book and market orders $Q = \min(Z, V)$ for $Z \sim \text{Pareto}(v_m, \gamma_m)$. Attempts that empty the ask book or do not move its best price are discarded before fitting $\hat{K}\Delta p = \ln Q$ through the origin. In the original argument, the exponential execution intensity is motivated by a market-order size density specified up to proportionality and a logarithmic price-impact relationship. We use this construction as a heuristic calibration, rather than as an identification of the execution intensity in our simulator. In particular, the normalized, capped Pareto distribution and random book depth do not imply the same exponential fill law. Moreover, the simulator permits partial and multi-unit executions over decision intervals, whereas the quote derivation assumes unit transactions. The resulting benchmark is therefore interpreted as an Avellaneda–Stoikov-inspired liquidation heuristic, without a claim of optimality under the simulated execution mechanism.

6.4 Empirical Results

6.4.1 Economic performance and auction use

The learned policies improve the trade-off between execution and terminal inventory relative to the AS and TWAP references. Table 4 summarizes ordinary and inventory-penalized shortfall; Figure 1 retains the market-level detail. Table 5 reports the paired reference comparisons and market-specific ordinary-shortfall levels. Every learner has lower inventory-penalized shortfall than both references in all six settings: all 48 paired improvement intervals are positive. These intervals are pointwise, without adjustment for multiple comparisons. This advantage does not extend to mean ordinary shortfall: all 24 learner–market means are higher than AS, as shown alongside the AS means in Table 5, although this is not a claim that every ordinary-shortfall difference is statistically established. Smaller inventory penalties account for their overall economic advantage.

No learner dominates the others. In the synthetic setting, mean penalized shortfall is between -4.5 and -4.0 bps. SAC has the lowest mean, but its paired comparisons with the other learners remain uncertain. MSFT is a useful counterexample to interpreting reference-relative gains as profitability: all four learners have positive penalized shortfall, with intervals above zero. Their mean ordinary shortfalls are also positive, although those intervals include zero (Table 5).

How much does the auction itself contribute? Holding a learner’s CLOB trajectory fixed, its auction contribution is the reduction in penalized shortfall relative to submitting no auction orders:

$$Z_{\tau^{\text{cl}}}(S_{\tau^{\text{cl}}}^{\text{cl}} - S_{\tau^{\text{op}}}^{\text{mid}}) - \sum_{i=n+1}^m d_{t_i} c_{t_i} + \lambda(I_{\tau^{\text{op}}}^2 - I_{\tau^{\text{cl}}}^2).$$

The execution-price edge net of fees reduces ordinary shortfall; the remaining term measures inventory-penalty relief, not cash income. Figure 2 shows positive mean contributions through-

Policy	Synthetic		Historical, equal-ticker mean	
	IS	IS $_{\lambda}$ [95% interval]	IS	IS $_{\lambda}$ [95% interval]
DQN	-5.11	-4.19 [-5.85, -2.76]	-7.12	-5.97 [-6.42, -5.52]
DDPG	-5.08	-4.03 [-5.97, -2.32]	-7.74	-6.68 [-7.04, -6.31]
TD3	-5.28	-4.40 [-6.25, -2.75]	-7.27	-6.39 [-6.70, -6.05]
SAC	-5.39	-4.54 [-6.35, -3.15]	-7.35	-6.44 [-6.86, -5.99]
AS	-6.45	-1.41 [-2.98, -0.06]	-8.44	-3.39 [-4.06, -2.71]
TWAP	-0.36	5.61 [4.10, 7.01]	-3.46	2.53 [2.27, 2.79]

Table 4: Headline shortfalls in initial-notional basis points; lower is better. Ordinary shortfall (IS) excludes the terminal inventory penalty, whereas IS $_{\lambda}$ includes it. Means weight ten seeds equally; historical summaries first average the five fixed tickers within seed. Brackets give pointwise 95% bootstrap intervals for penalized-shortfall levels only, not paired policy differences. Historical order flow and auctions are simulated on reused price dates.

out the headline settings. Synthetic TD3 is the only case whose interval includes zero, at 0.35 bps with a 95% interval of $[-0.02, 0.73]$. The continuous learners obtain larger mean contributions than DQN in both the synthetic and pooled, fixed-ticker historical summaries. In the latter, most of the contribution comes from reducing inventory risk rather than improving execution prices. These comparisons follow each learner’s own CLOB policy; they do not measure auction skill at a common opening inventory.

The policies sell most inventory before the call, but auction trading need not reduce absolute holdings. Synthetic DQN, TD3 and SAC increase mean absolute inventory while reducing its mean quadratic penalty. The two observations are compatible: absolute inventory and squared inventory weight large positions differently. Any residual position is marked at the frozen midprice, not forcibly executed.

Retraining with and without auction access asks a different question, because the CLOB policy can adapt to the available closing liquidity. In this matched synthetic comparison (E-0 versus No-A; Table 7 and the bottom panel of Figure 3), auction access lowers penalized shortfall for all four learners, with positive paired intervals. Thus the evidence supports useful auction access in these settings without implying complete liquidation or a benefit outside the tested policies and calibration.

6.4.2 Forecast information

Table 6 displays the forecast errors. The projected clearing signal has lower mean absolute error (MAE) and root mean squared error (RMSE) than the contemporaneous midprice in both phases of every setting. The improvement is modest during continuous trading and more pronounced during the call. On synthetic paths, auction mean absolute error falls from about 0.147 to 0.050 model-price units. This measures prediction of the simulator’s clearing price, not an observed exchange auction.

The estimand is forecast error along the realized paths of the selected deterministic headline DQN, DDPG, TD3 and SAC policies, evaluated on the 100 test episodes per algorithm, master seed and market specified in Section 6.1. We record H_t^{cl} and S_t^{mid} before each action at decision times $t < \tau^{\text{cl}}$, including the initial CLOB decision and auction opening. Terminal observations at τ^{cl} , where the

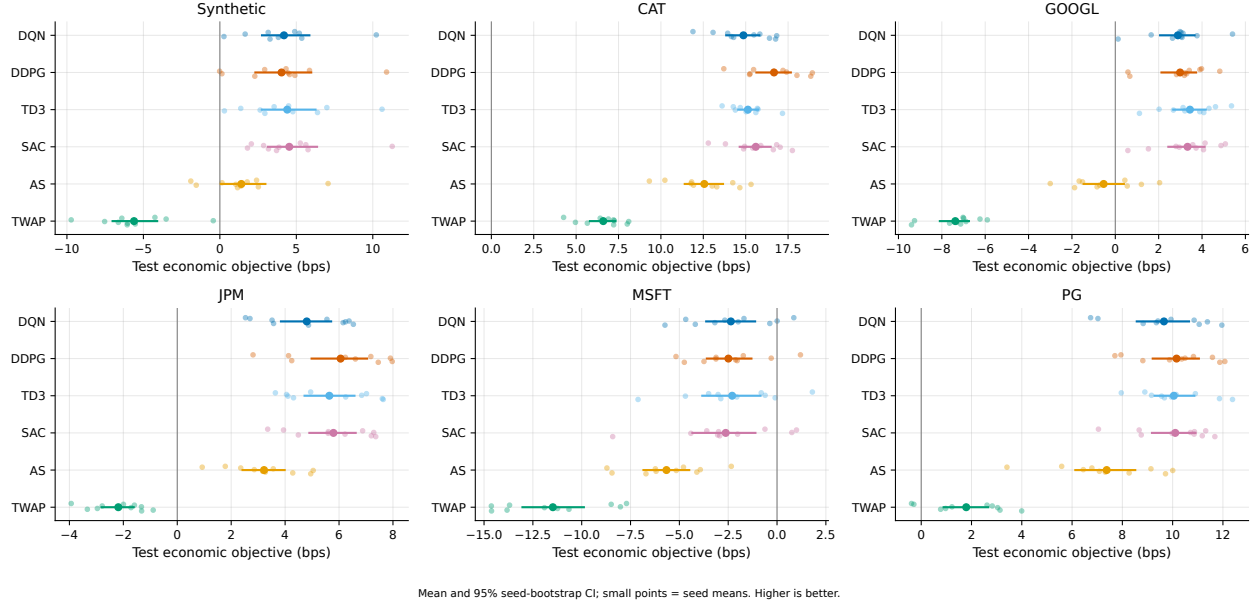


Figure 1: Headline economic performance by market, reported as the economic score $-\text{IS}_\lambda$ in bps, so higher values mean lower penalized shortfall. Large points and bars show means and 95% intervals; small points show seed means. AS and TWAP are reference policies. Horizontal scales differ across markets.

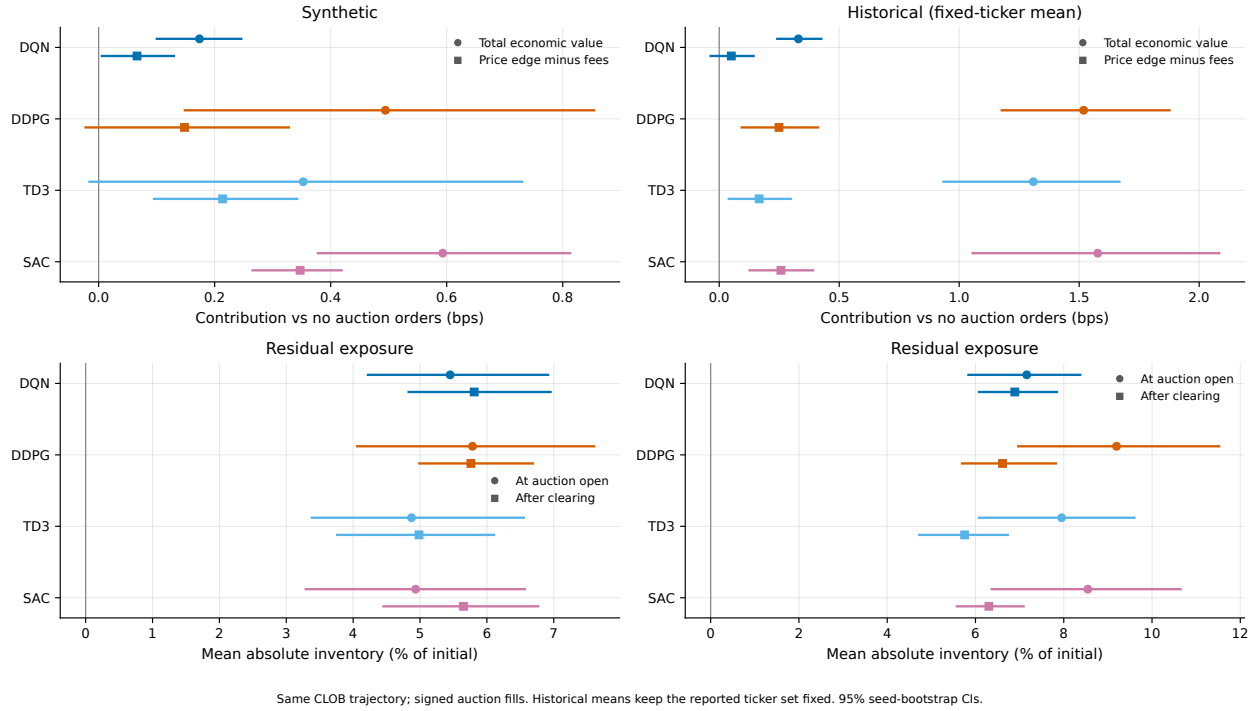


Figure 2: Auction contribution and inventory management on each policy's CLOB trajectory. Top: total economic value is penalized-shortfall reduction; price edge minus fees is ordinary-shortfall reduction. Bottom: mean absolute inventory at opening and clearing, as a percentage of initial inventory. Historical panels average the five tickers equally. Positive contributions favor participation; bars are 95% intervals. Retrained access is reported separately in Figure 3.

Market	Policy	Ordinary shortfall	Paired penalized-shortfall reduction	
			IS [95% CI]	versus AS [95% CI] versus TWAP [95% CI]
Synthetic	DQN	-5.11 [-6.76, -3.76]	2.79 [2.25, 3.42]	9.81 [8.92, 10.59]
	DDPG	-5.08 [-7.04, -3.27]	2.63 [1.70, 3.50]	9.65 [8.72, 10.53]
	TD3	-5.28 [-7.15, -3.58]	3.00 [2.14, 3.92]	10.02 [9.11, 10.81]
	SAC	-5.39 [-7.30, -3.95]	3.13 [2.69, 3.56]	10.15 [9.32, 11.03]
	AS	-6.45 [-8.27, -4.98]	—	—
CAT	DQN	-16.18 [-16.98, -15.30]	2.31 [1.80, 2.86]	8.27 [7.71, 8.78]
	DDPG	-17.60 [-18.60, -16.51]	4.11 [3.33, 4.90]	10.07 [9.65, 10.54]
	TD3	-15.76 [-16.40, -15.13]	2.57 [1.58, 3.44]	8.53 [7.79, 9.23]
	SAC	-16.32 [-17.14, -15.41]	3.04 [2.39, 3.72]	9.00 [8.38, 9.62]
	AS	-17.61 [-18.39, -16.69]	—	—
GOOGL	DQN	-4.10 [-4.82, -3.20]	3.42 [2.84, 4.01]	10.26 [9.90, 10.64]
	DDPG	-4.10 [-4.85, -3.25]	3.53 [2.87, 4.23]	10.37 [10.10, 10.63]
	TD3	-4.39 [-5.18, -3.49]	3.98 [3.10, 4.82]	10.82 [10.26, 11.37]
	SAC	-4.23 [-4.99, -3.30]	3.87 [2.77, 4.95]	10.71 [10.13, 11.28]
	AS	-4.51 [-4.96, -3.96]	—	—
JPM	DQN	-5.64 [-6.54, -4.71]	1.58 [0.80, 2.38]	6.99 [6.27, 7.68]
	DDPG	-6.95 [-8.05, -5.78]	2.84 [2.03, 3.68]	8.24 [7.58, 8.82]
	TD3	-6.47 [-7.49, -5.47]	2.42 [1.37, 3.36]	7.83 [7.02, 8.55]
	SAC	-6.64 [-7.54, -5.72]	2.57 [1.82, 3.34]	7.98 [7.39, 8.49]
	AS	-8.26 [-8.94, -7.53]	—	—
MSFT	DQN	0.87 [-0.52, 2.34]	3.30 [2.69, 3.76]	9.11 [8.29, 9.97]
	DDPG	0.92 [-0.26, 2.02]	3.17 [2.64, 3.70]	8.99 [8.23, 9.76]
	TD3	1.05 [-0.42, 2.65]	3.36 [2.16, 4.40]	9.18 [8.23, 10.13]
	SAC	1.37 [-0.24, 3.06]	3.03 [1.50, 4.42]	8.85 [7.82, 9.89]
	AS	0.61 [-0.60, 1.76]	—	—
PG	DQN	-10.54 [-11.41, -9.55]	2.28 [1.79, 2.77]	7.86 [7.59, 8.11]
	DDPG	-10.95 [-11.89, -9.92]	2.78 [1.99, 3.49]	8.37 [8.01, 8.71]
	TD3	-10.76 [-11.58, -9.93]	2.67 [1.74, 3.48]	8.26 [7.74, 8.72]
	SAC	-10.92 [-11.74, -9.96]	2.73 [1.73, 3.72]	8.31 [7.78, 8.87]
	AS	-12.42 [-13.26, -11.51]	—	—

Table 5: Market-specific shortfalls in basis points. Ordinary shortfall reports policy levels; paired reductions equal reference-minus-learner penalized shortfall, so positive values favor the learner. Brackets give pointwise 95% seed-bootstrap intervals, unadjusted for multiple comparisons.

indicative price equals settlement, are excluded. The auction midprice comparator is frozen at opening. Each forecast is compared with that episode’s eventual simulated clearing price under the same policy, which can depend on its subsequent actions; this is not a no-order or policy-independent forecasting target.

Within each phase and episode, average absolute errors and squared errors over its pre-action decision times. Average these episode means equally over the 100 episodes for each algorithm and seed, then equally over the four algorithms within seed and the ten seeds. MAE is the resulting mean absolute error; RMSE is the square root of the final averaged squared error, not an average of episode RMSEs. This episode-first construction gives equal episode weight despite random CLOB

decision counts; it does not pool all timestamps across episodes. It follows the bundled analysis protocol and seed-level summaries, rather than the separate time-bin diagnostics. This evaluation is distinct from the 256 no-order training paths used to fit reliability weights in Section 2.1.4.

Market	Phase	Mean absolute error (MAE)				RMSE	
		H	Midprice	Reduction	[95% CI]	H	Midprice
Synthetic	CLOB	0.1726	0.1837	0.0111	[0.0097, 0.0126]	0.2775	0.2831
	Auction	0.0504	0.1471	0.0967	[0.0884, 0.1050]	0.0908	0.1964
CAT	CLOB	0.1127	0.1314	0.0187	[0.0177, 0.0197]	0.1675	0.1811
	Auction	0.0331	0.1200	0.0868	[0.0813, 0.0921]	0.0505	0.1598
GOOGL	CLOB	0.0806	0.0989	0.0183	[0.0177, 0.0188]	0.1132	0.1313
	Auction	0.0267	0.1025	0.0758	[0.0712, 0.0814]	0.0454	0.1308
JPM	CLOB	0.1169	0.1348	0.0179	[0.0165, 0.0194]	0.1586	0.1720
	Auction	0.0345	0.1470	0.1124	[0.1082, 0.1164]	0.0532	0.1702
MSFT	CLOB	0.1273	0.1485	0.0212	[0.0203, 0.0221]	0.1859	0.2077
	Auction	0.0413	0.1579	0.1166	[0.1084, 0.1243]	0.0680	0.1869
PG	CLOB	0.0752	0.0863	0.0111	[0.0106, 0.0116]	0.1096	0.1176
	Auction	0.0229	0.0963	0.0734	[0.0683, 0.0783]	0.0353	0.1163

Table 6: Forecast accuracy by market and trading phase, in model-price units. H is the projected clearing forecast; Midprice is the contemporaneous midprice, held fixed at auction opening during the call. Lower MAE and root mean squared error (RMSE) indicate greater accuracy. A positive MAE reduction (Midprice minus H) favors H; brackets give its pointwise 95% confidence interval.

The incremental decision value of the clearing forecast depends on the learner (Table 7; Figure 3). With the frozen-mid anchor, economic objective and dense conditioning held fixed, explicit access to H worsens DQN’s penalized shortfall, while the effects for DDPG, TD3 and SAC remain uncertain. SAC achieves the lowest mean synthetic penalized shortfall under the headline specification and shows a positive estimated benefit from H access in this matched information comparison. Together, these findings suggest that SAC can use clearing information effectively, although neither its superiority over the other learners nor its incremental H-access benefit is conclusively established: the paired learner comparisons remain uncertain, and SAC’s H-access interval narrowly includes zero. The two findings concern different specifications: the headline uses the full training scheme with indicative anchoring, whereas the information comparison uses economic training with frozen-mid anchoring. H-off retains other clearing-relevant observations, so these results do not distinguish redundant information from a learner’s difficulty exploiting the forecast.

The clearing forecast provides a natural reference around which the agent learns quote deviations, but accurate prediction alone does not guarantee an effective liquidation strategy. The agent receives H as its estimate of the clearing price; inventory, execution probability, fees and price impact also shape its preferred action. Relative to the frozen-mid anchor, indicative anchoring worsens penalized shortfall for DDPG, TD3 and SAC; the DQN effect is uncertain. Both conditions retain H access, so the comparison concerns how this information is used to parameterize quotes. The two anchors expose different parts of the bounded action grid, affecting both the available quote levels and the learning problem. Thus belief-based anchoring is an intuitive design choice that can underperform the tested alternative; this comparison does not identify a globally optimal strategy. Forecast

accuracy, access to the H feature and its use as a quote anchor are therefore separate questions.

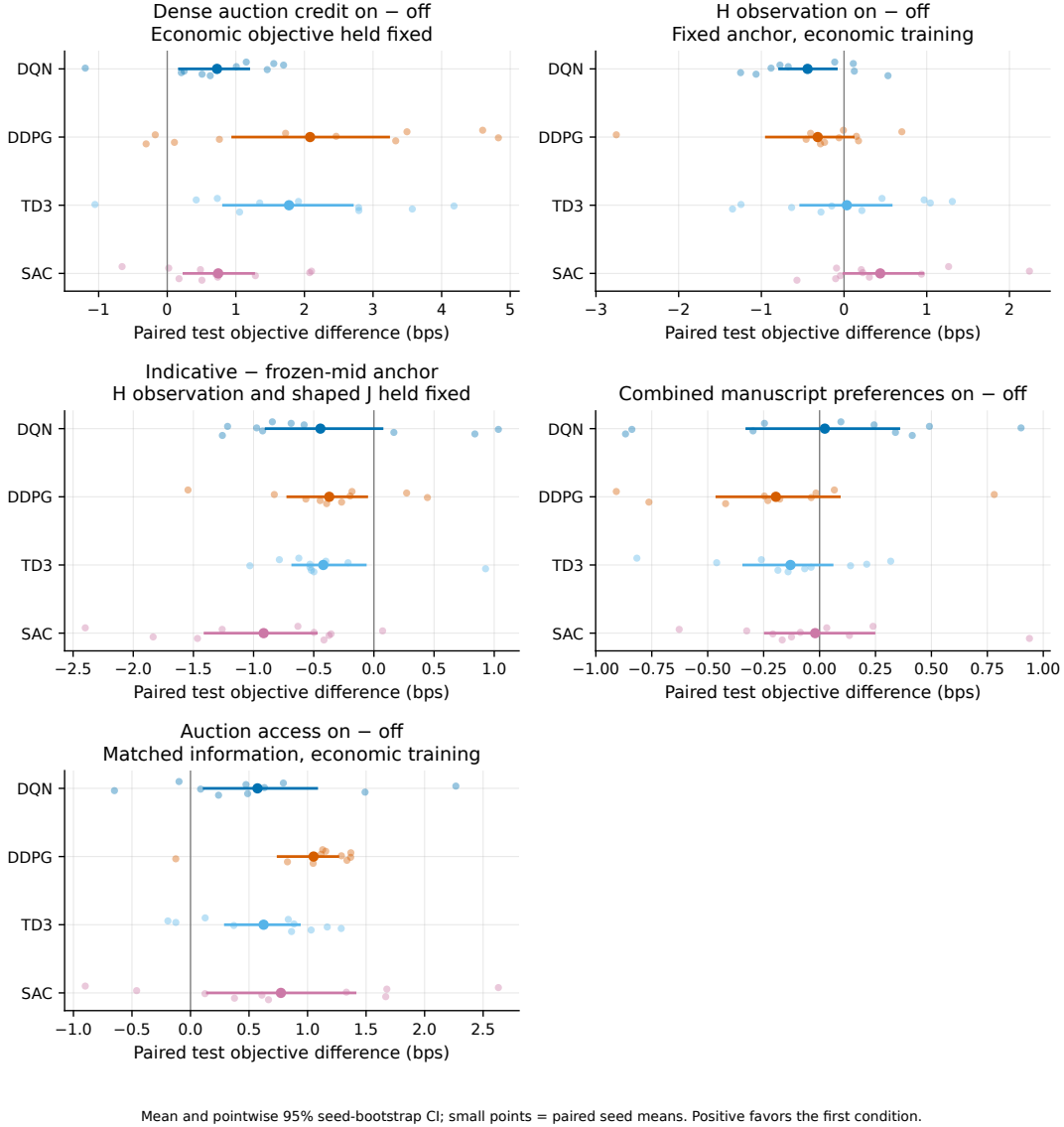


Figure 3: Matched synthetic controls: dense versus sparse auction credit, explicit H access, quote anchoring, combined preferences and auction access (read across rows). Economic differences equal comparator-minus-proposed penalized-shortfall reductions in bps. Positive values favor the first condition. Large points and bars show means and 95% intervals; small points show paired seed means.

6.4.3 Learning from delayed auction outcomes

Because auction orders settle only at the close, a learner receives little immediate economic feedback about its decisions. Dense auction credit helps bridge that delay. Against sparse credit, it reduces selected-policy test penalized shortfall for all four learners, with mean improvements ranging from 0.73 to 2.08 bps and positive intervals throughout. The validation comparisons also favor dense credit for every learner at episode 250 (Figure 4). This comparison changes the auction potential

while retaining the economic objective and common CLOB conditioning; it does not add an economic cash flow.

The complete headline training scheme also outperforms raw cash-flow training on average. At the indicative anchor, the mean reduction in penalized shortfall is 3.53–5.16 bps across learners, with positive intervals for all four methods. The headline scheme wins 39 of the 40 paired seed means; DQN at seed 314 is the exception, with penalized shortfall higher by 0.15 bps. This is a broader comparison than dense versus sparse credit: raw cash-flow training removes both the preferences in J and learning conditioning, including initial-value centering. The gains cannot be assigned to any one of those changes.

The matched preference comparisons show no established marginal benefit from the additional preferences. At the indicative anchor, adding the forecast-relative preferences while retaining conditioning worsens SAC penalized shortfall by 0.16 bps, with a 95% interval of $[0.05, 0.30]$ for the increase; the other learners remain uncertain. At the frozen-mid anchor, every preference-effect interval includes zero. Dense credit therefore has a demonstrated benefit in these experiments, whereas the additional preferences can be detrimental. Uncertain effects neither prove equivalence nor identify the effects of individual preference terms.

Table 7 collects the matched effects, separating the broad training-scheme comparison from changes to individual design choices.

Contrast (first versus second)	DQN	DDPG	TD3	SAC
Dense versus sparse credit	0.73	2.08	1.78	0.74
E-F versus E-S	[0.18, 1.19]	[0.95, 3.23]	[0.82, 2.70]	[0.24, 1.26]
H access	−0.44	−0.32	0.03	0.44
E-F versus E-0	[−0.78, −0.09]	[−0.94, 0.11]	[−0.52, 0.57]	[0.00, 0.95]
Indicative versus frozen-mid anchor	−0.44	−0.37	−0.42	−0.92
J-I versus J-F	[−0.90, 0.07]	[−0.72, −0.06]	[−0.67, −0.07]	[−1.40, −0.48]
Preferences, frozen-mid anchor	0.02	−0.20	−0.13	−0.02
J-F versus E-F	[−0.32, 0.35]	[−0.46, 0.09]	[−0.34, 0.06]	[−0.24, 0.24]
Retrained auction access	0.57	1.05	0.62	0.77
E-0 versus No-A	[0.12, 1.08]	[0.75, 1.26]	[0.30, 0.93]	[0.14, 1.41]
Complete scheme versus raw cash flow	3.53	3.65	4.83	5.16
J-I versus Cash-I	[1.10, 6.64]	[2.84, 4.57]	[4.03, 5.70]	[4.08, 6.27]
Preferences, indicative anchor	−0.01	0.04	−0.12	−0.16
J-I versus E-I	[−0.60, 0.53]	[−0.60, 0.57]	[−0.31, 0.06]	[−0.30, −0.05]

Table 7: Matched synthetic effects: second-condition minus first-condition penalized shortfall in basis points, so positive values favor the first condition. Each cell reports a mean reduction and its pointwise 95% percentile interval over ten paired seed means. Schemes are defined in Table 2; the final two rows are the indicative-anchor follow-ups. Both preference contrasts hold conditioning fixed; the raw-cash-flow contrast removes both preferences and conditioning. Intervals are not adjusted for multiple comparisons.

These controlled comparisons concern the synthetic market at the stated calibration. The historical results use a simulated order-flow and auction mechanism on previously inspected price paths, and the algorithm comparisons include the implementation differences described in Section 4. Together, the results support auction participation and dense feedback within this model, not real-exchange

profitability or global policy optimality.

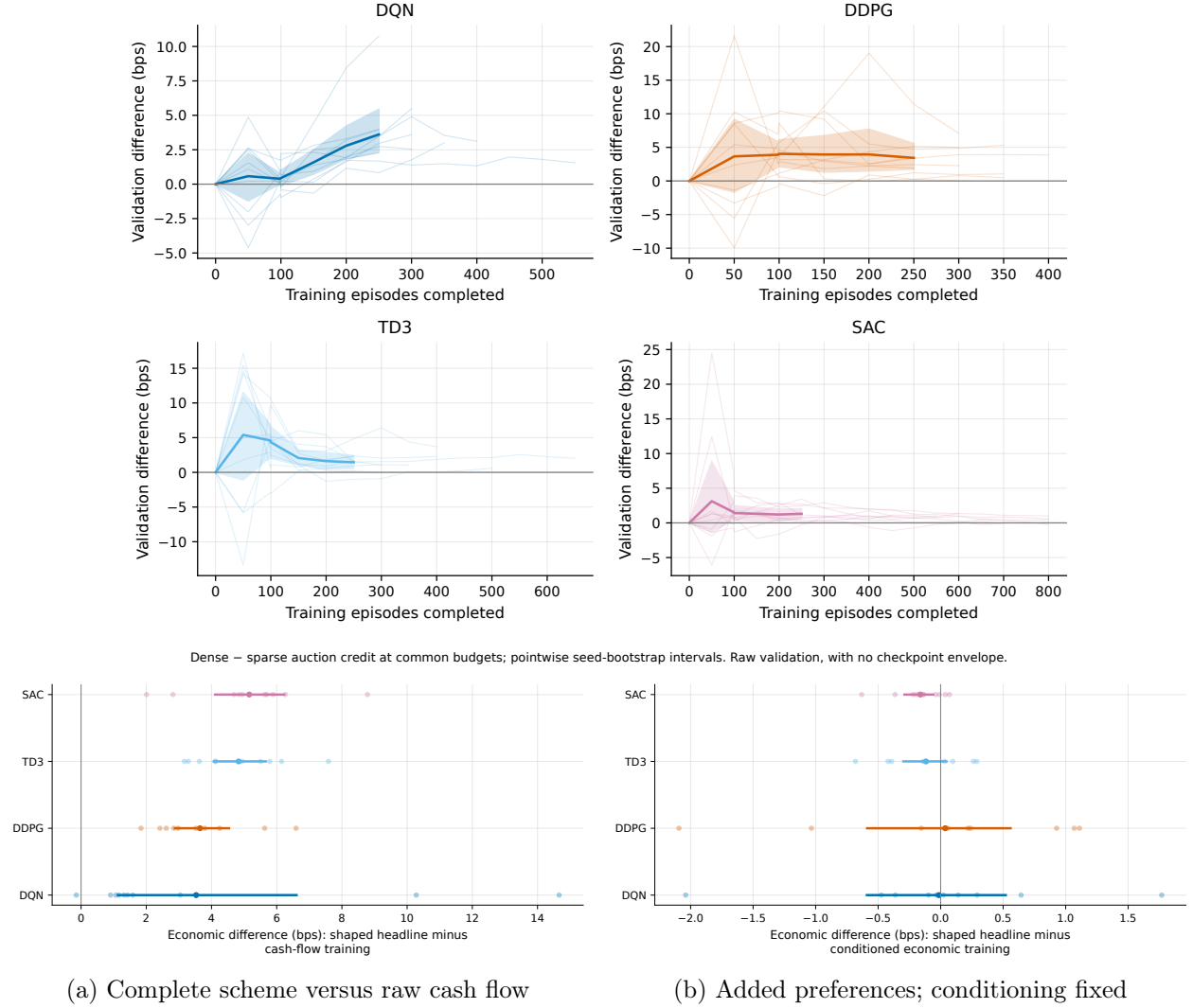


Figure 4: Learning from delayed auction outcomes. Top: validation gains from dense rather than sparse credit (E-F versus E-S); thin lines show individual seeds and bands show 95% intervals at common training budgets. Bottom: selected-policy test gains from the complete scheme over raw cash-flow training (left) and from added preferences with conditioning fixed (right). All reductions concern penalized shortfall, in bps; positive values favor dense credit or the headline scheme.

Figure 5 shows how validation performance varies across seeds and where the selection procedure stops. It complements the selected-policy test comparisons rather than supplying additional test evidence.

7 Conclusion

A closing auction is both a source of execution liquidity and a final opportunity to manage inventory. In our experiments, policies that anticipate the call improve inventory-penalized shortfall relative to AS and TWAP, while their mean ordinary shortfalls remain higher than AS. The main finding

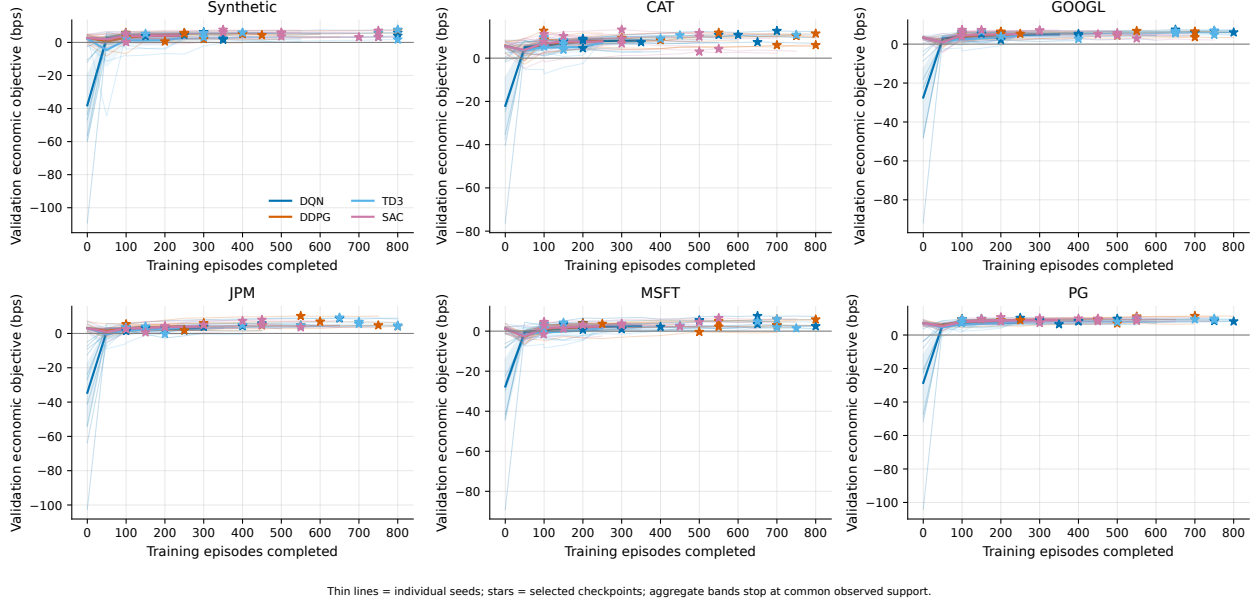


Figure 5: Validation economic score $-\text{IS}_\lambda$ (bps; higher is better). Thin lines show seed trajectories; stars mark selected checkpoints. Heavy lines and 95% bands stop at the last budget observed for all seeds, while individual traces may continue. Episode zero is the untrained model. No smoothing or extrapolation is applied.

is that anticipating the auction can improve the execution–inventory trade-off, with dense feedback helping learners use this opportunity. Historical auction contributions come mainly from inventory-risk relief. Retraining with auction access also benefits all four learners in the synthetic setting. Dense auction credit improves learning despite the delay before settlement. The forecast itself is informative, but its incremental decision value depends on the learner, and using it as a quote anchor can worsen performance. Once learning conditioning is held fixed, we find no established marginal benefit from the added forecast-relative preferences, and they worsen SAC performance at the indicative anchor.

These findings suggest that RL has the potential to be effective for optimal liquidation and *in fine* market making in complex structures beyond simple LOBs with closing auction, as for example workup session or AHEAD mechanism [22, 19] or sequence of periodic auctions [49, 43, 11].

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